Interstellar Dust
and
Radiative Process

CHEM

1D UCLCHEM

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Leiden University/Observatory – The Netherlands

UCLCHEM Workshop – Bonn 2026

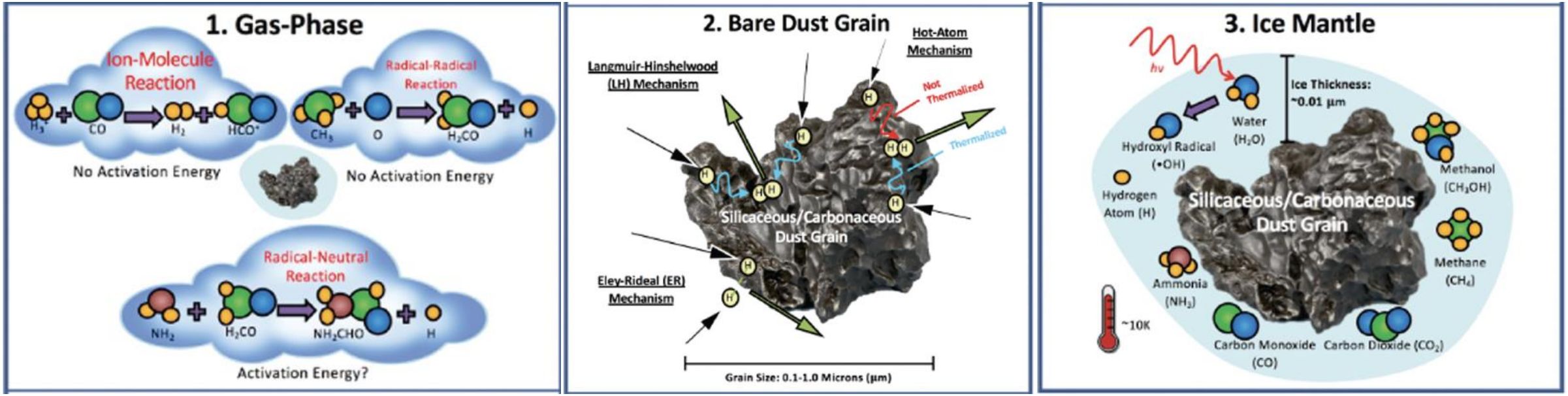
COMs are Ubiquitous

Molecules in the Interstellar Medium or Circumstellar Shells (as of 03/2026)

2 atoms	3 atoms	4 atoms	5 atoms	6 atoms	7 atoms	8 atoms	9 atoms	10 atoms	11 atoms	12 atoms	>12 atoms
CH^+	H_2O	NH_3	HC_3N	CH_3OH	CH_3CHO	$HC(O)OCH_3$	$(CH_3)_2O$	$(CH_3)_2CO$	HC_9N	c-C_6H_6*	C_{60}*
CH	HCO^+	H_2CO	HCOOH	CH_3CN	CH_3C_2H	CH_3C_3N	CH_3CH_2OH	$(CH_2OH)_2$	CH_3C_6H	n-C_3H_7CN	C_{70}*
CN	HCN	HNCO	H_2CNH	NH_2CHO	CH_3NH_2	C_7H	CH_3CH_2CN	CH_3CH_2CHO	C_2H_5OCHO	i-C_3H_7CN	C_{60}^{+}*
OH	OCS	H_2CS	H_2NCN	CH_3SH	CH_2CHCN	CH_3COOH	HC_7N	CH_3C_5N	$CH_3OC(O)CH_3$	$C_2H_5OCH_3$	c-C_6H_5CN
CO	HNC	C_2H_2*	H_2C_2O	C_2H_4*	HC_5N	C_6H_2	CH_3C_4H	CH_3CHCH_2O	$CH_3C(O)CH_2OH$	1-c-C_5H_5CN	$HC_{11}N$
H_2	H_2S	C_3N	C_4H	C_5H	C_6H	CH_2OHCHO	C_8H	CH_3OCH_2OH	c-C_5H_6	2-c-C_5H_5CN	1-$C_{10}H_7CN$
SiO	N_2H^+	HNCS	SiH_4*	CH_3NC	c-C_2H_4O	l-HC_6H*	$CH_3C(O)NH_2$	c-C_6H_4	$HOCH_2CH_2NH_2$	CH_3C_7N (?)	2-$C_{10}H_7CN$
CS	C_2H	$HOCO^+$	c-C_3H_2	HCCCHO	H_2CCHOH	CH_2CHCHO	C_8H^-	H_2CCCHC_3N	H_2CCCHC_4H	n-C_3H_7OH	c-C_9H_8
SO	SO_2	C_3O	H_2CCN	l-C_4H_2	C_6H^-	CH_2CCHCN	C_3H_6	C_2H_5NCO	$C_{10}H^-$	i-C_3H_7OH	1-c-C_5H_5CCH
SiS	HCO	l-C_3H	C_5*	HC_3NH^+	CH_3NCO	H_2NCH_2CN	CH_3CH_2SH	$C_2H_5NH_2$ (?)	$H_2C(CH)_3CN$?	$(CH_3)_2C{=}CH_2$	2-c-C_5H_5CCH
NS	HNO	$HCNH^+$	SiC_4	C_5N	HC_5O	CH_3CHNH	CH_3NHCHO	HC_7NH^+	$CH_3CH(OH)CHO$ (2026)		c-$C_5H_4CCH_2$
C_2**	HCS^+	H_3O^+	l-C_3H_2	l-HC_4H*	$HOCH_2CN$	CH_3SiH_3	HC_7O	E-$CH_3CHCHCN$	CH_3OCH_2CHO (2026)		2-C_9H_7CN
NO	HOC^+	C_3S	CH_4*	l-HC_4N	HCCCHNH	$H_2NC(O)NH_2$	HCCCHCHCN	Z-$CH_3CHCHCN$			C_6H_5CCH
HCl	c-SiC_2	c-C_3H	HCCNC	c-H_2C_3O	HC_4NC	$HCCCH_2CN$	H_2CCHC_3N	$CH_3C(CN)CH_2$			$CH_3OCH_2CH_2OH$ (2024)
NaCl	C_2S	HCCN	HNC_3	H_2CCNH	c-C_3HCCH	HC_5NH^+	$H_2CCCHCCH$	CH_2CHCH_2CN			1-$C_{12}H_7CN$ (2024)
KCl	C_3*	H_2CN	H_2COH^+	C_5N^-	l-H_2C_5	CH_2CHCCH	HOCHCHCHO (2024)	$HOCH_2C(O)NH_2$			5-$C_{12}H_7CN$ (2024)
AlCl	CO_2*	c-SiC_3	C_4H^-	HNCHCN	MgC_5N	MgC_6H	HC_7N^+ (2024)	CH_3CH_2CCH (2024)			1-$C_{16}H_9CN$ (2024)
AlF	CH_2	CH_3*	HC(O)CN	SiH_3CN	CH_2C_3N	$C_2H_3NH_2$	$CH_2(CCH)_2$ (2024				2-$C_{16}H_9CN$ (2025)
PN	C_2O	C_3N^-	HNCNH	C_5S	NC_4NH^+	$(CHOH)_2$	$(CH_3)_2S$ (2025)				4-$C_{16}H_9CN$ (2025)
SiC	MgNC	PH_3	CH_3O	MgC_4H	MgC_5N^+	$HC_2(H)C_4$					$C_{24}H_{11}CN$ (2025)
CP	NH_2	HCNO	NH_4^+	CH_3CO^+	HC_5N^+ (2024)	C_7N^-					$C_{13}H_{10}$ (2025)
NH	NaCN	HOCN	H_2NCO^+	C_3H_3	HNC_5 (2024)	CH_3CHCO					3-$C_{12}H_7CN$ (2026)
SiN	N_2O	HSCN	$NCCNH^+$	H_2C_3S	$CH_2(CN)_2$ (2024	MgC_6H^+					4-$C_{12}H_7CN$ (2026)
SO^+	MgCN	H_2O_2	CH_3Cl	HCCCHS	HCCCHCN (2025)	Z-$(CH)_2(CN)_2$ (2024					c-$C_{11}H_8$ (2026)
CO^+	$H_3^{+(*)}$	C_3H^+	MgC_3N	C_5O	CH_3CHS (2025)	CH_2CHCHS (2025)					
HF	SiCN	HMgNC	NH_2OH	C_5H^+	SiC_6 (2025)						
SiH?	AlNC	HCCO	HC_3O^+	$HCCNCH^+$							
FeO?	SiNC	CNCN	HC_3S^+	c-C_3C_2H							

https://cdms.astro.uni-koeln.de/classic/molecules

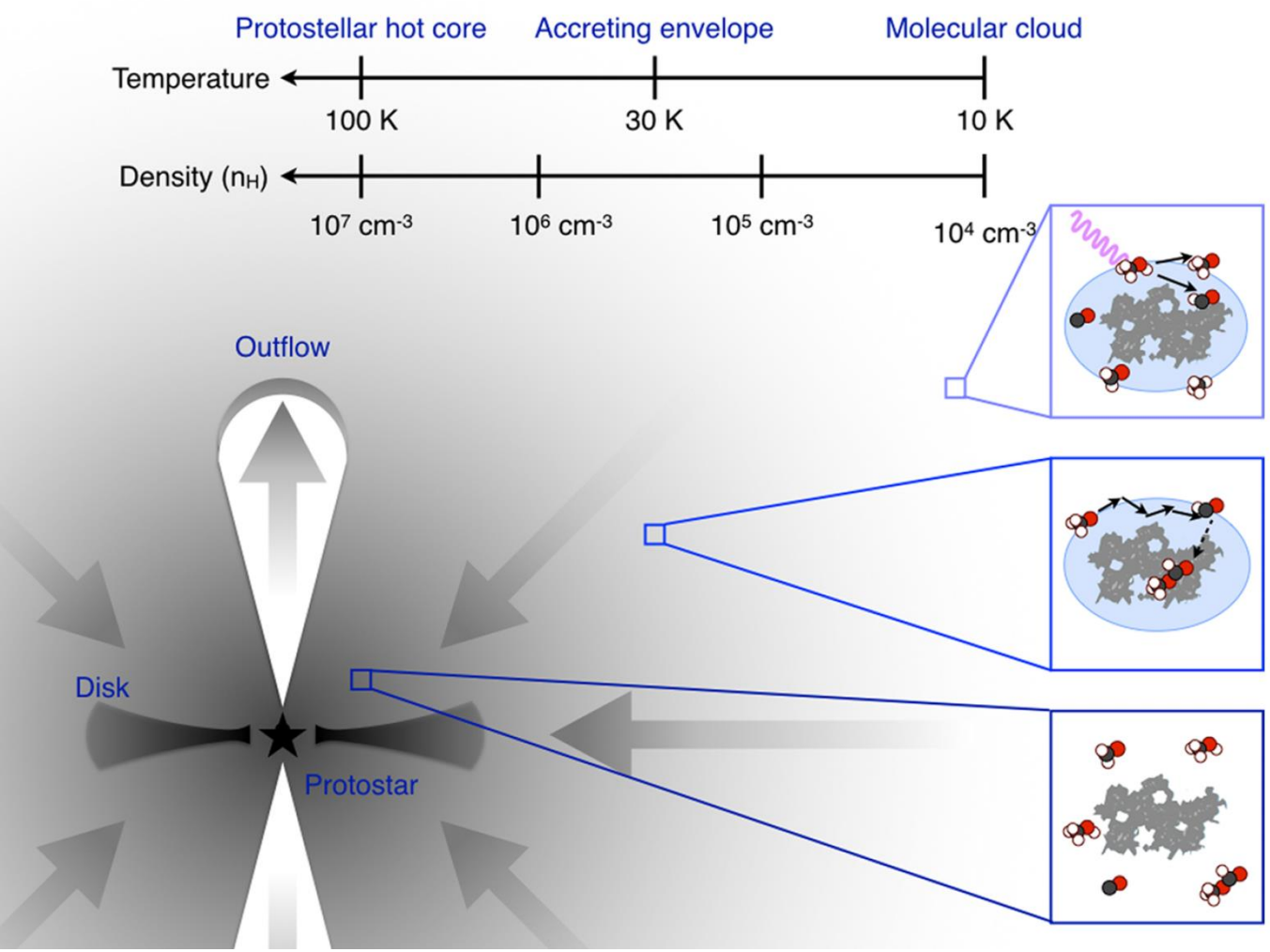
Interstellar Dust as a Key Contributor



Arumainayagam et al. 2019 (Chem Soc Rev)

- Gas-phase reactions alone are not efficient enough
- Dust acts like heterogeneous catalysis
- Cosmic ices is the most important environment for interstellar chemical evolution

COMs are associated with warm ISM

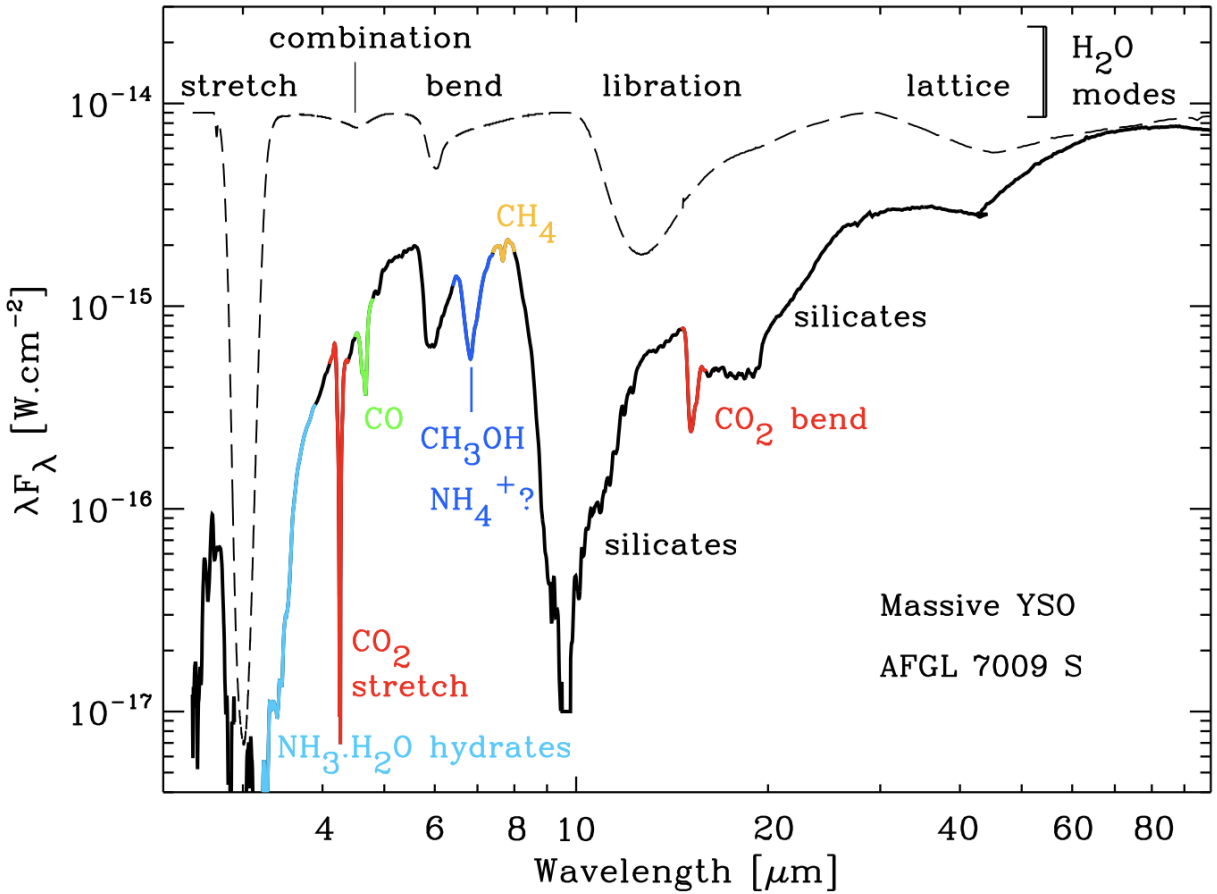


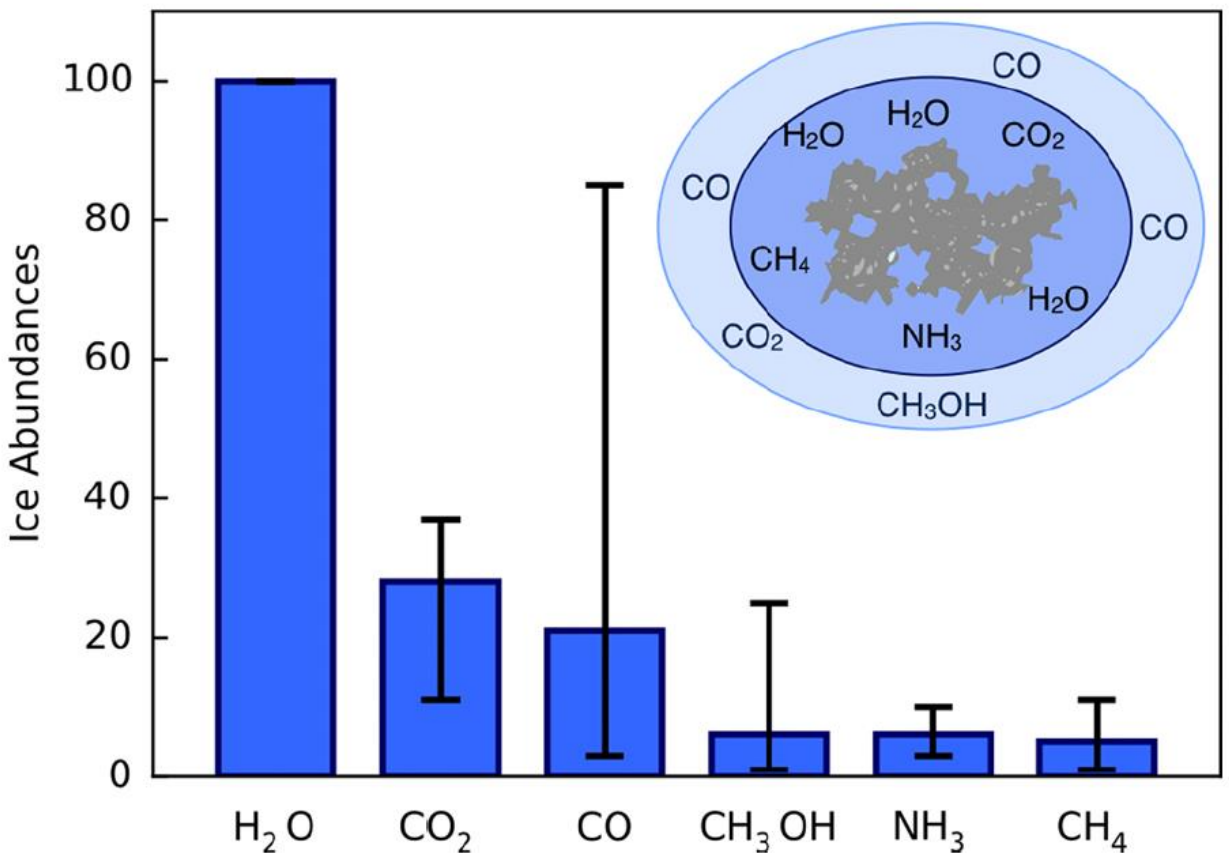
1. Simple ices formed at low-Temp
2. Ices + UV → radicals
3. Icy grain heated up
   → mobility of radicals
   → form more complex species
4. Hot enough → thermal desorption

Öberg 2016 (Chem Rev)

Interstellar Ice Compositions

YSO AFGL 7009 with ISO

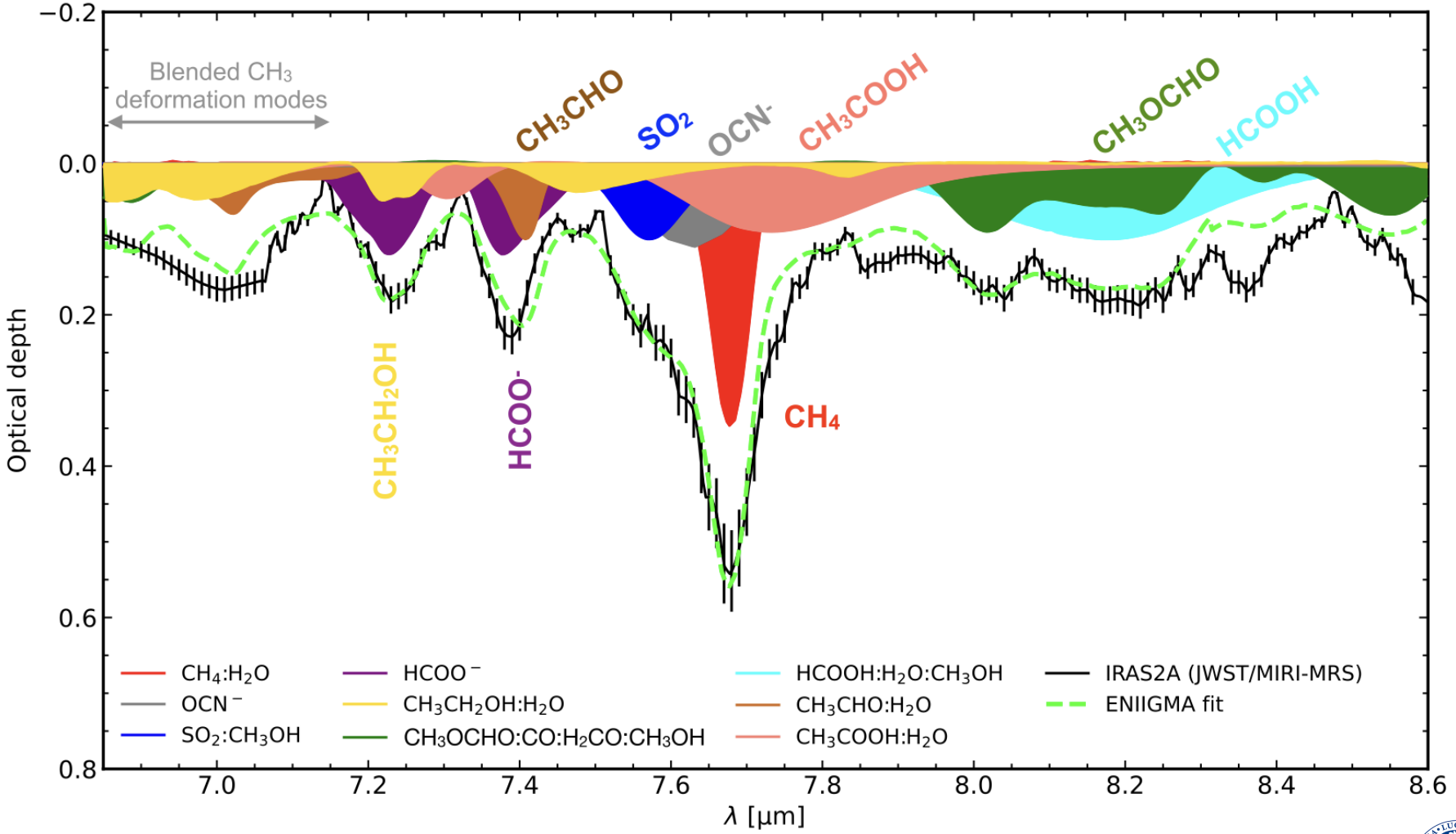




Boogert et al. 2015 (ARAA)

Interstellar Ice Compositions

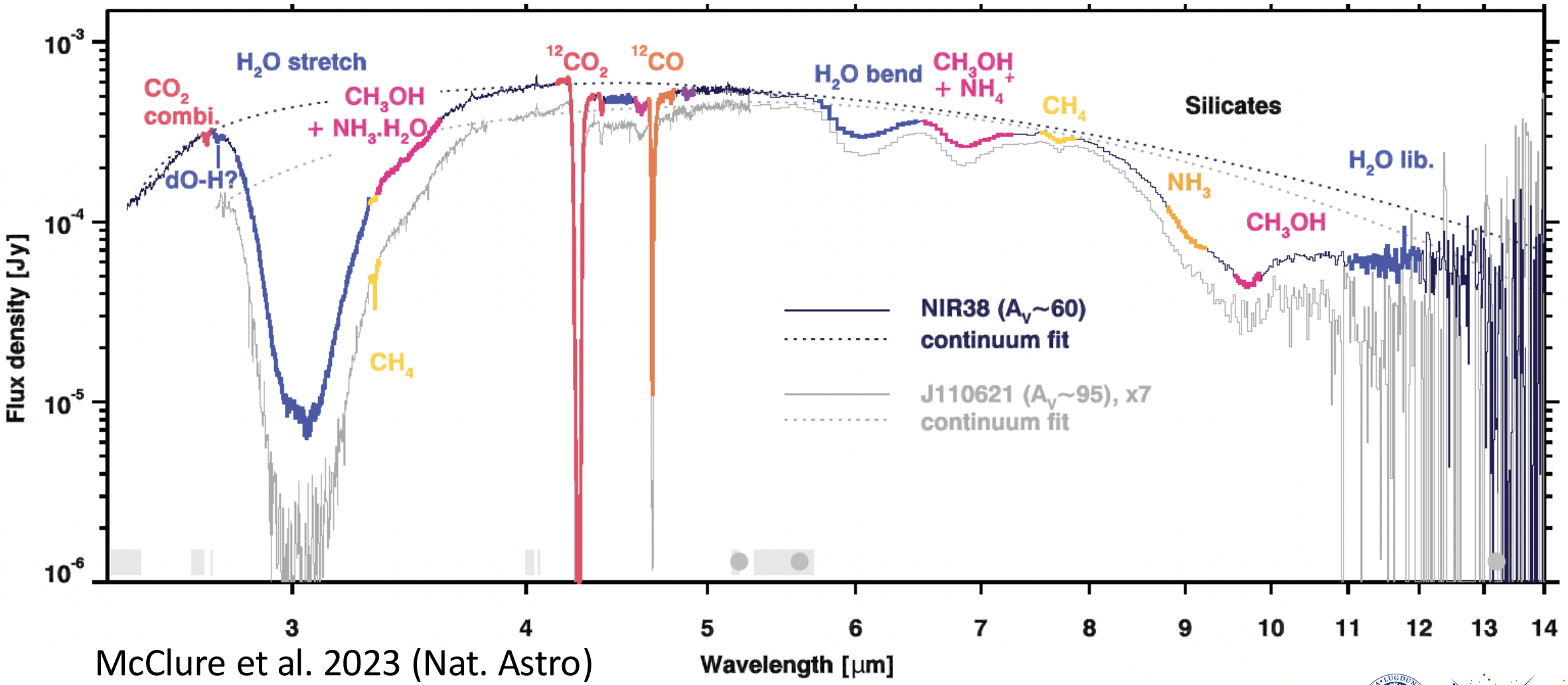
High-mass protostar (IRAS 2A) with JWST



Rocha et al. 2024 (A&A)

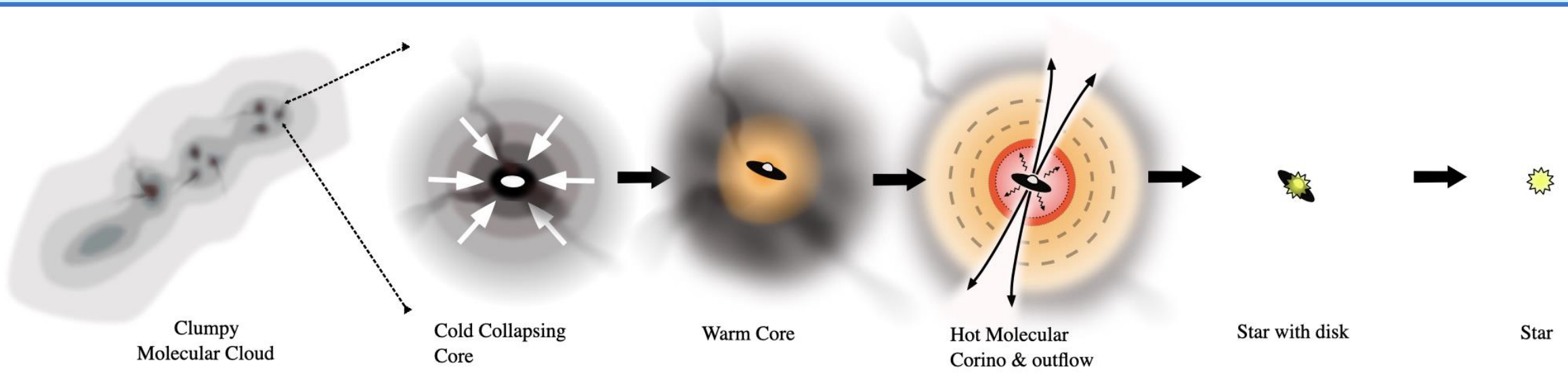
Interstellar Ice Compositions

Dark Clouds in Chameleon with JWST

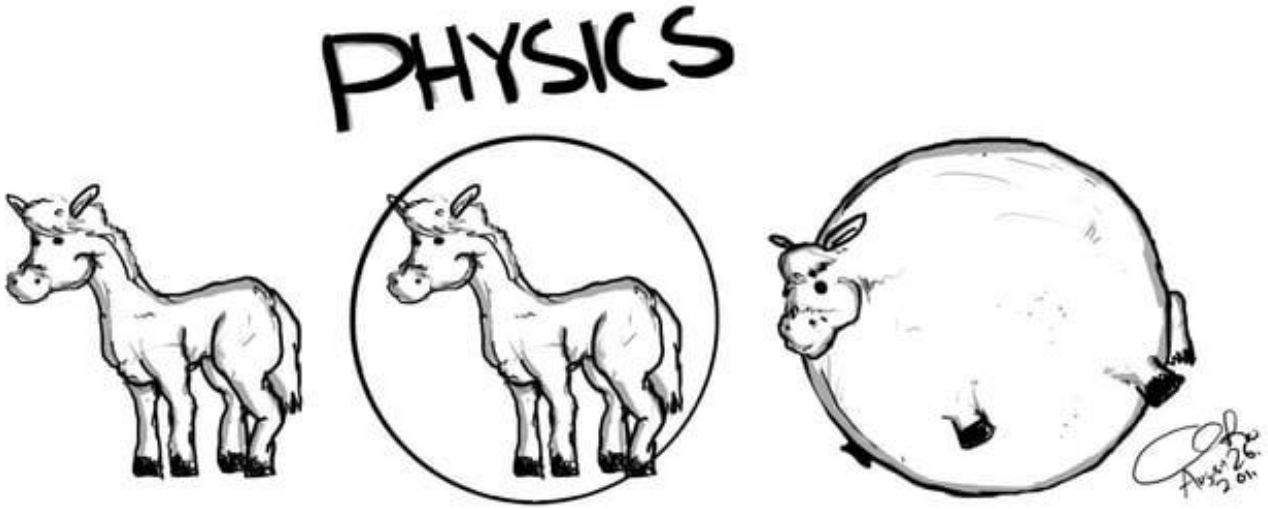


McClure et al. 2023 (Nat. Astro)

Formation of Protostellar Cores

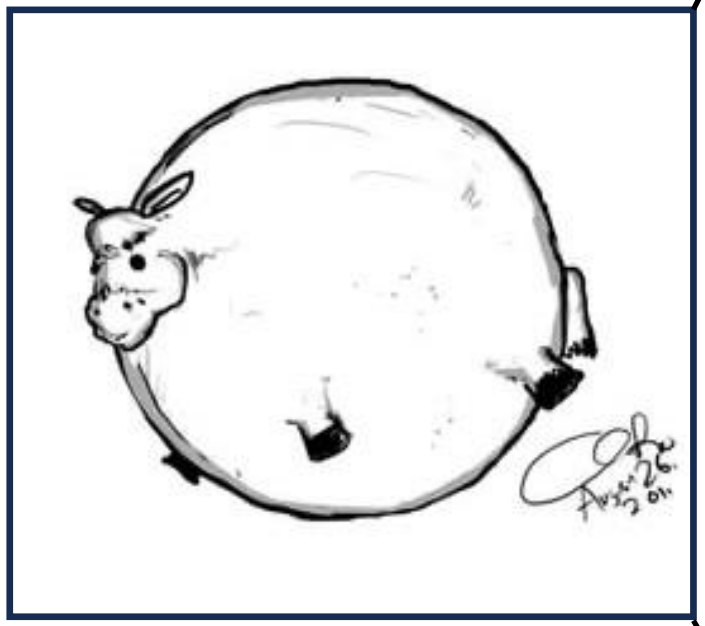


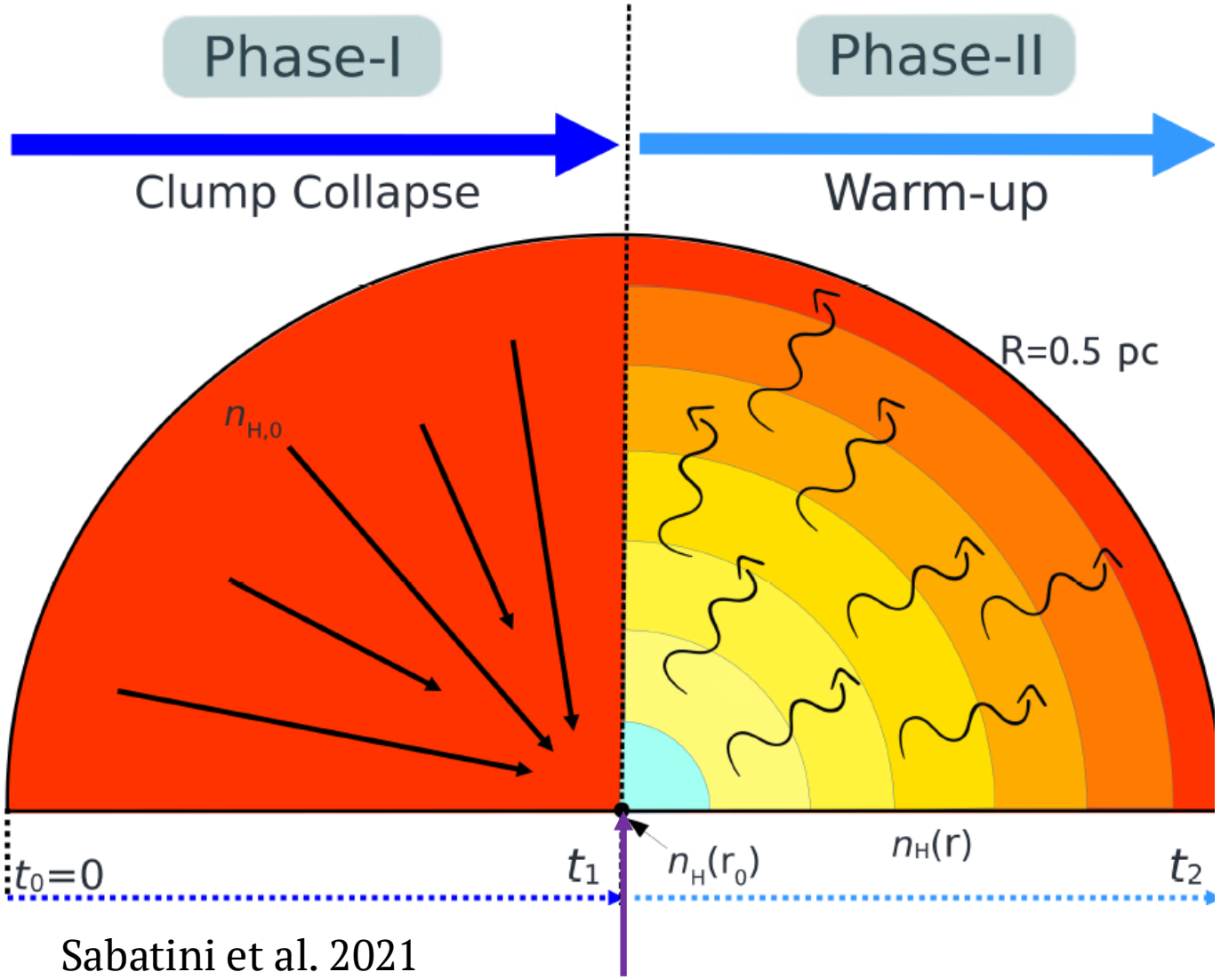
original cartoon by Cormac Purcell, modified by A. Ginsburg



Credit: https://www.reddit.com/r/Physics/comments/jwjqo/the_animal_is_too_complicated_for_the_math_make/

Collapsing + Heating Phases



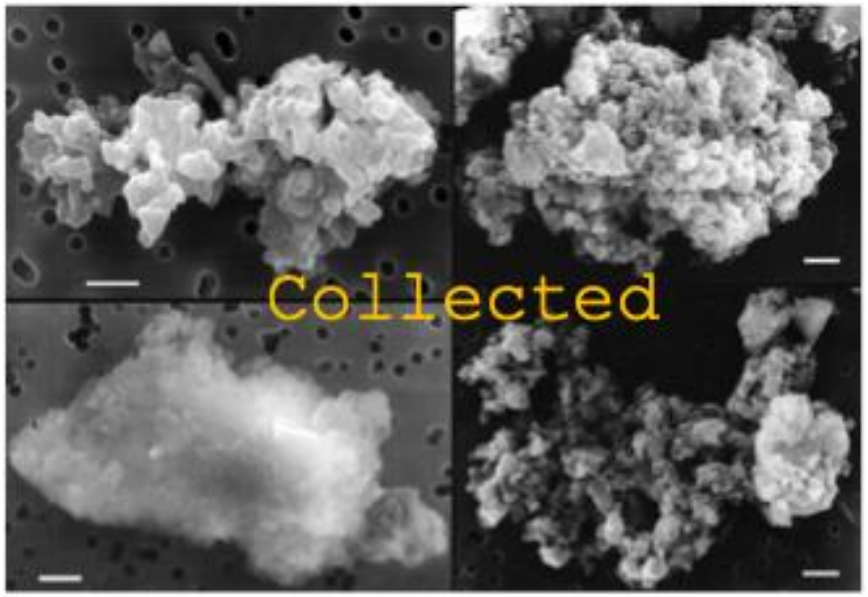


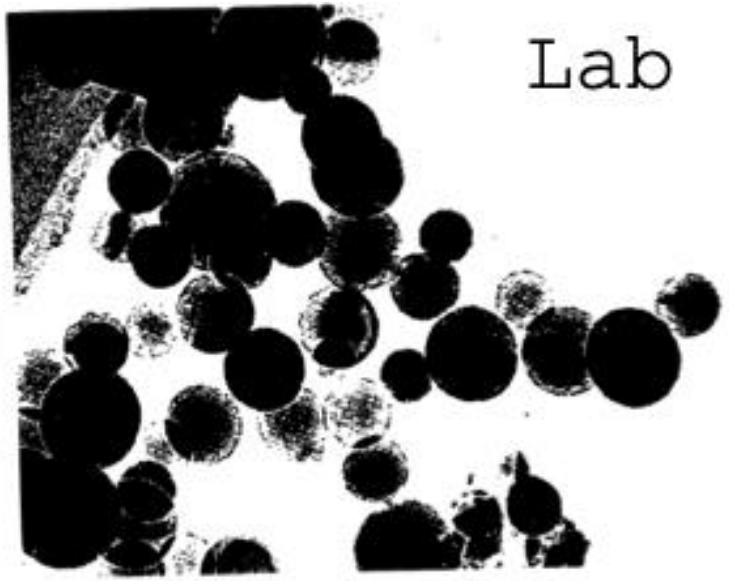
Sabatini et al. 2021

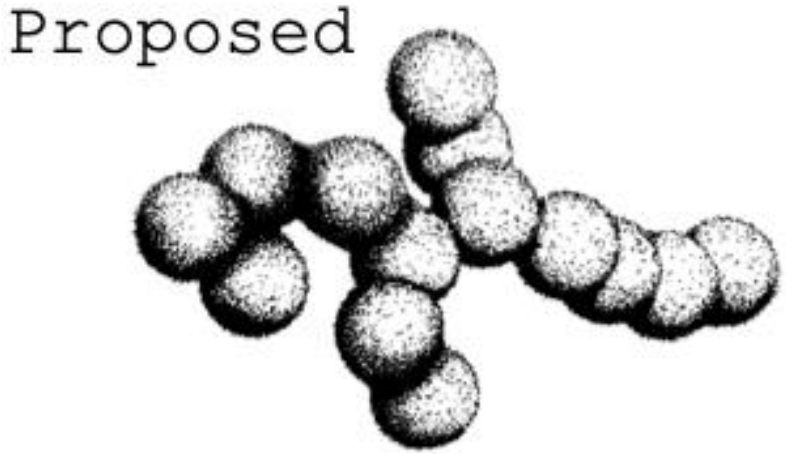
Formation of protstellar core

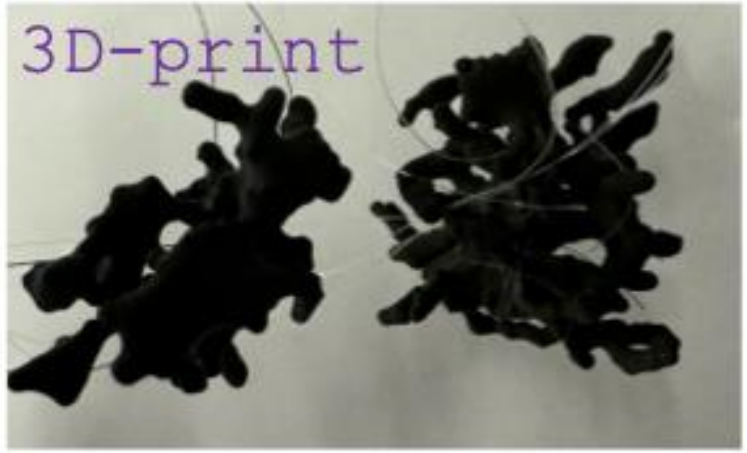
No outflow
No Clumpiness

Intrinsic Properties of ISM Dust



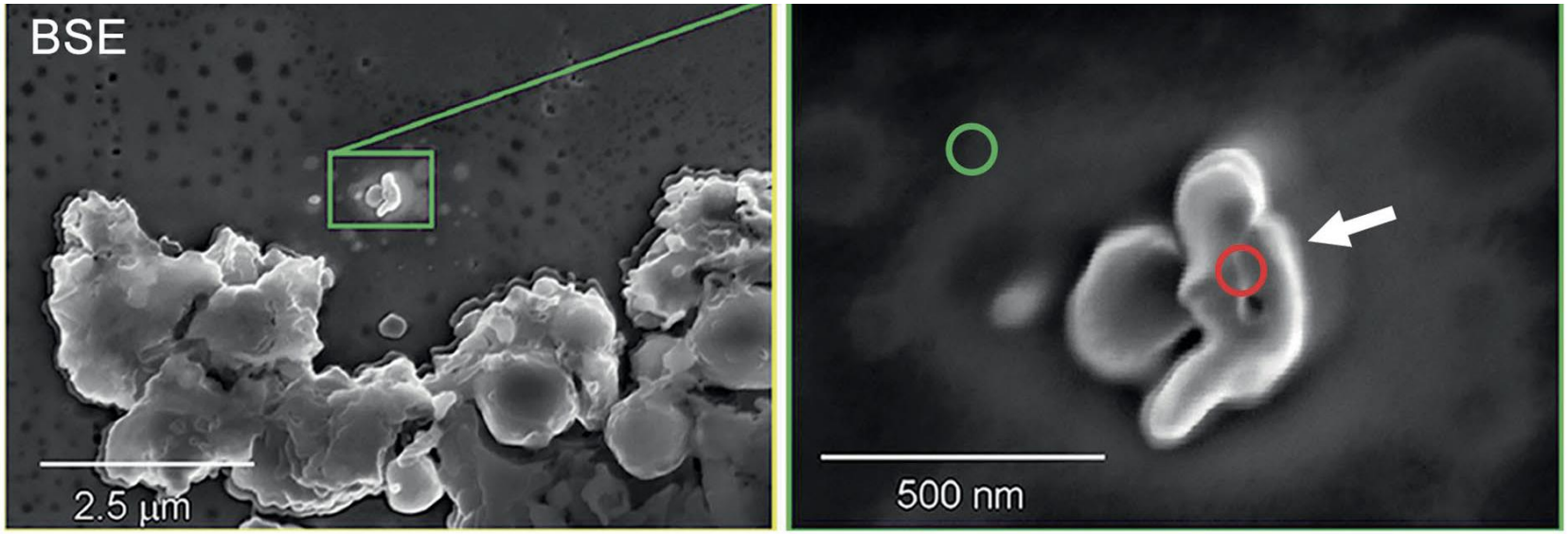


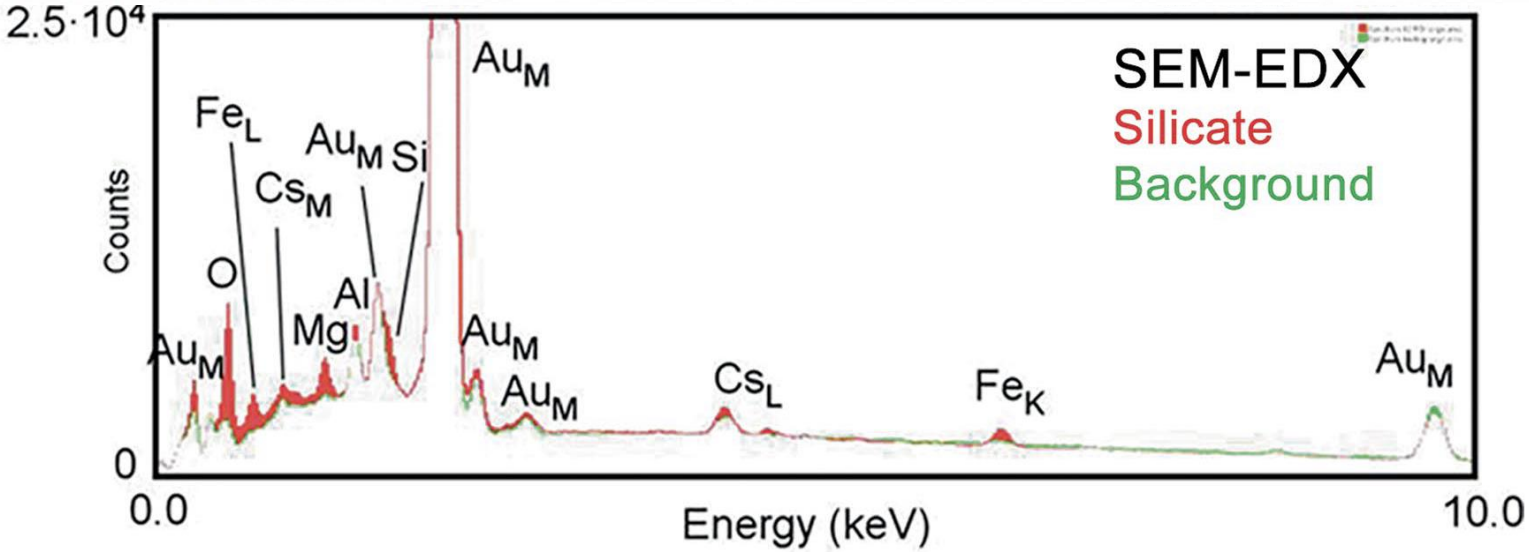




Intrinsic Properties of ISM Dust

Samples of asteroid Bennu
NASA's OSIRIS-REx mission





Barnes et al. 2025 (Nat. Astro)

Diagnostics of Interstellar Dust

- ❑ Emission
- ❑ Extinction and Reddening Effect
- ❑ Scattering polarization

→ Size and mineralogy

- ❑ Absorption/Emission Polarization

Non-spherical grain

→ Size, shape, structure, mineralogy

Interactions between Dust and Radiation Field

- **Spherical grain** with same volume
- No size-distribution

- **Mie theory** to characterize interactions with radiation (electromagnetic wave)

  (Absorption, Scattering and Extinction cross-sections)

Gustav Mie

Dust Temperature at Equilibrium

- Absorption rate of energy

$$\frac{dE_{\rm abs}}{dt} = \int_\nu \pi a^2 F_\nu Q_{\rm abs}(a,\nu) d\nu$$

  - From radiation intensity to flux $F_\nu = \int_\Omega I_\nu(\Omega) cos\theta d\Omega = \pi I_\nu$
  - From radiation intensity to energy density $u_\lambda = \frac{1}{c}\int_\Omega I_\nu(\Omega) d\Omega = \frac{4\pi I_\nu}{c}$

- Final Bbsorption rate

$$\frac{dE_{\rm abs}}{dt} = \pi a^2 \int_\nu c u_\nu Q_{\rm abs}(a,\nu) d\nu$$

NOTE: u_λ is a function of A_V

Dust Temperature at Equilibrium

- Emission rate of energy

$$\frac{dE_{\rm emi}}{dt} = \int_\nu [\pi B_\nu(T_{\rm dust})][\pi a^2 Q_{\rm abs}] d\nu$$

$$\frac{dE_{\rm emi}}{dt} = 4\pi^2 a^2 \langle Q_{\rm abs}(a,\nu)\rangle_{\rm IR} \frac{\sigma_{\rm SB} T_{\rm dust}^4}{\pi}$$

$$\langle Q_{\rm abs}(a,\nu)\rangle_{\rm IR} = \frac{\int_\nu B_\nu(T_{\rm dust}) Q_{\rm abs}(a,\nu) d\nu}{\int_\nu B_\nu(T_{\rm dust}) d\nu}$$

$$\langle Q_{\rm abs}(a,\nu)\rangle_{\rm IR} = q_{\rm abs} T_{\rm dust}^2 a$$

- Final emission rate

$$\frac{dE_{\rm emi}}{dt} = 4\pi a^3 q_{\rm abs} \sigma_{\rm SB} T_{\rm dust}^6$$

Dust Temperature at Equilibrium

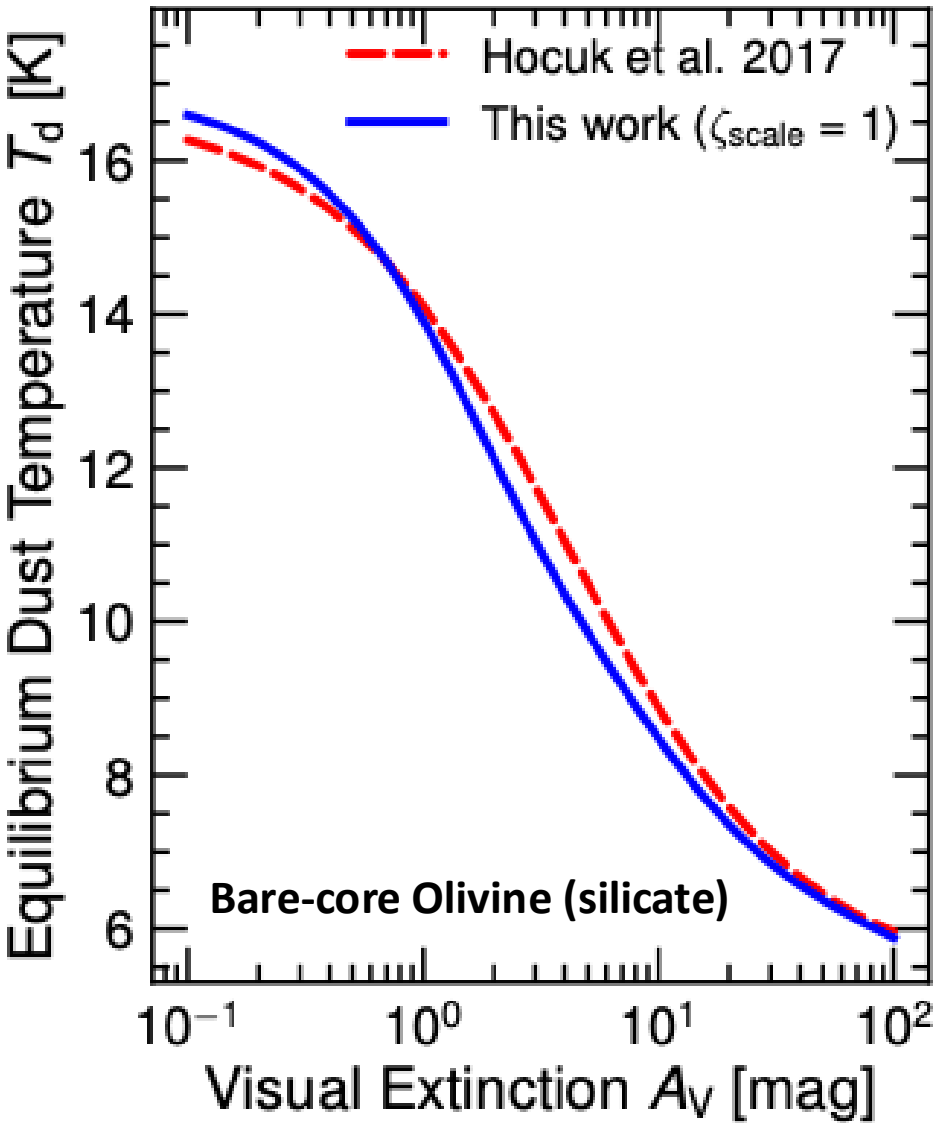
- Equilibrium

$$\frac{dE_{\rm abs}}{dt} \equiv \frac{dE_{\rm emi}}{dt}$$

- Empirical fit (Hocuk et al. 2017)

$$T_{\rm d}^{\rm Hoc} = [11 + 5.7 \times \tanh(0.61 - \log_{10}(A_{\rm V}))]\, \chi_{\rm uv}^{1/5.9},$$

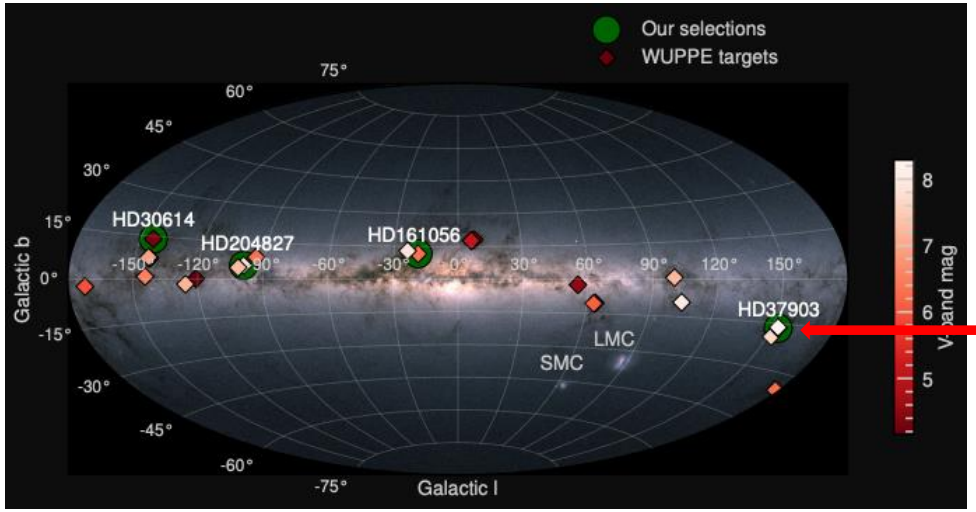
Dust heating by ISRF



Optical Depth (not spectral-line optical depth)

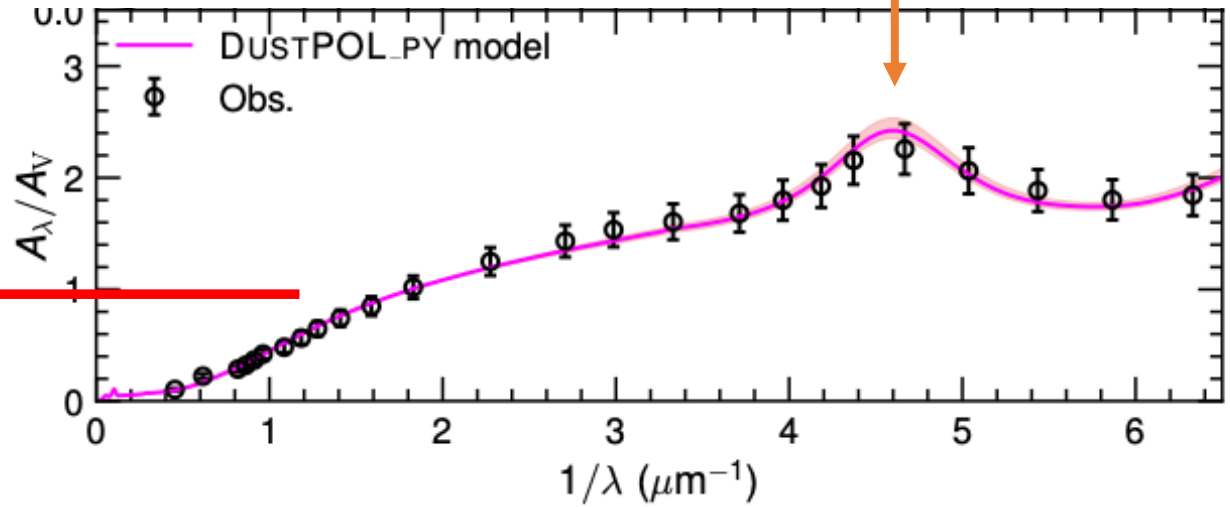
- Visual extinction vs. optical depth $A_\lambda = 1.086\tau_\lambda$

- Extinction curve A_λ/A_V can be modelled/calculated (forward modelling)



Tram et al. (subm)

2175 Å feature ← PAHs



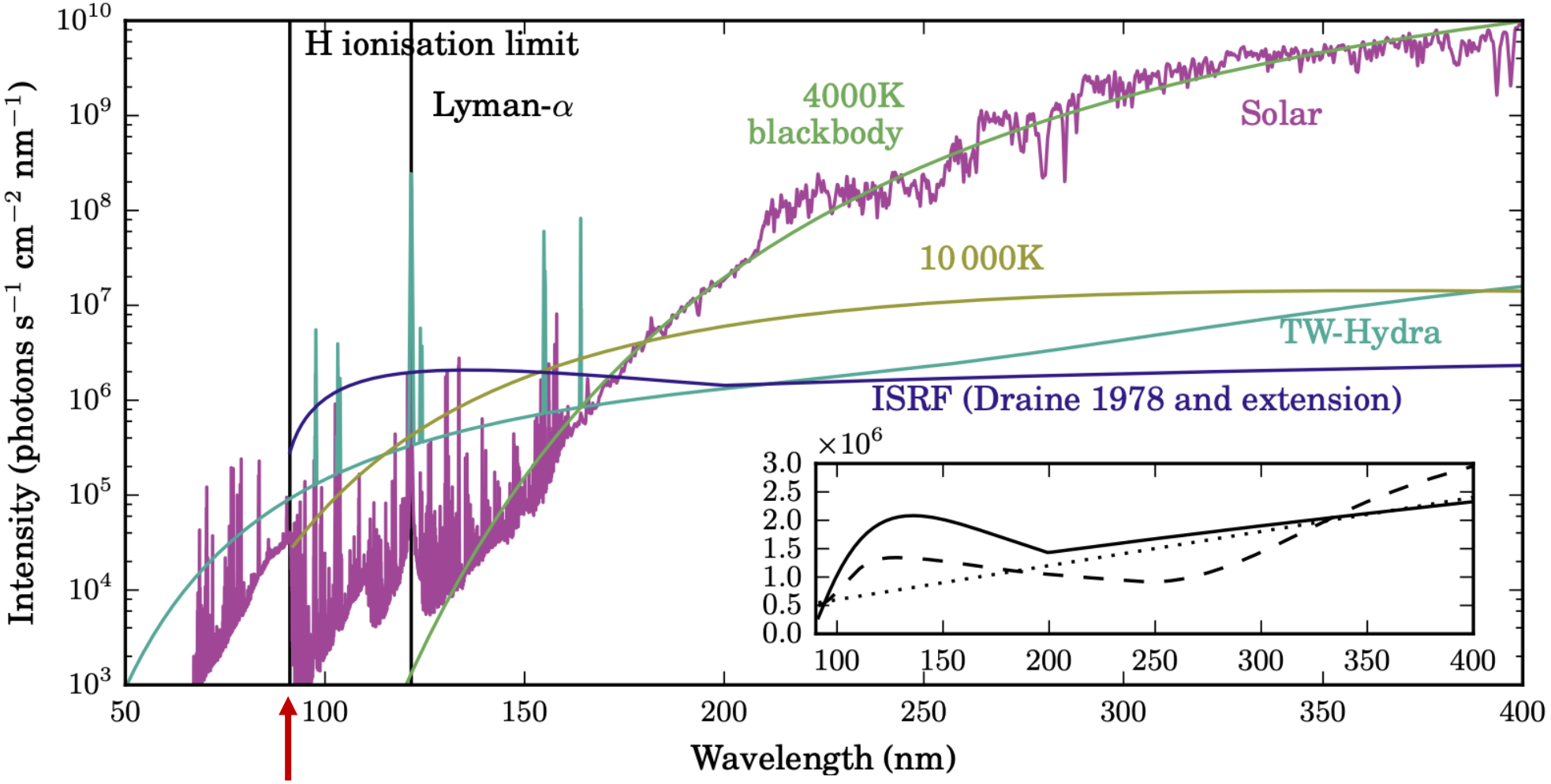
Optical Depth (not spectral-line optical depth)

- Visual extinction vs. optical depth $A_\lambda = 1.086\tau_\lambda$

- Extinction curve can be modelled/calculated (forward modelling)

- Data-driven curve A_λ/A_V (reverse model) ← local/LoS-dependent
  - CCM fit (Cardelli et al. 1989) ($1.1 < \lambda^{-1} < 3.3\ \mu\mathrm{m}^{-1}$) **(adopted in UCLCHEM)**
  - FM fit (Fitzpatrick & Massa, 1990, …) ($\lambda^{-1} > 3.3\ \mu\mathrm{m}^{-1}$)
  - Improvement and calibration carried out by community (e.g. Karl D. Gordon (STScI) with large surveys)

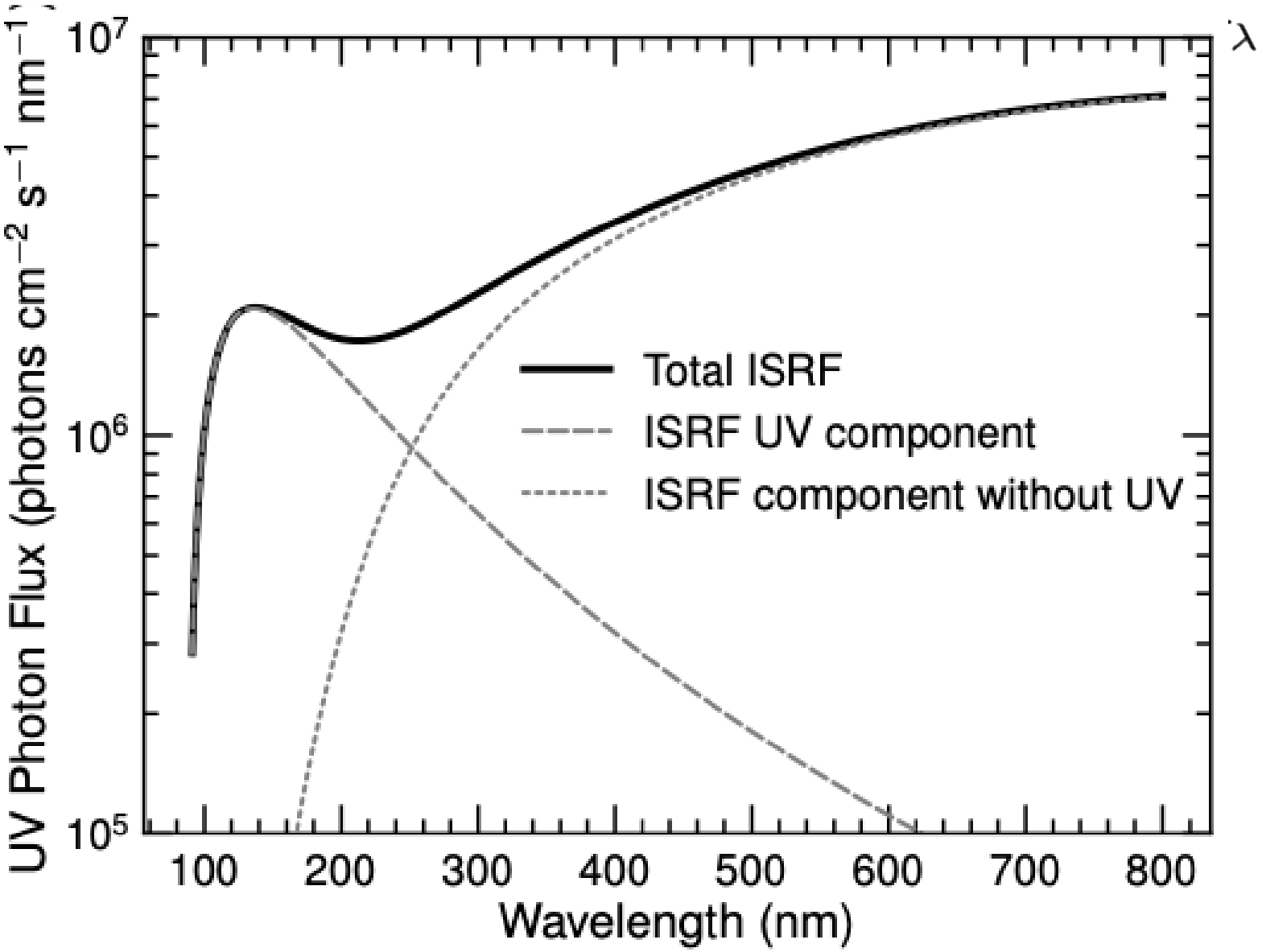
Radiation Fields



13.6 eV

https://home.strw.leidenuniv.nl/~ewine/photo/radiation_fields.html

Interstellar Radiation Field (ISRF)



Draine's field

Exercise to compute T_{dust}

❑ What was provided?

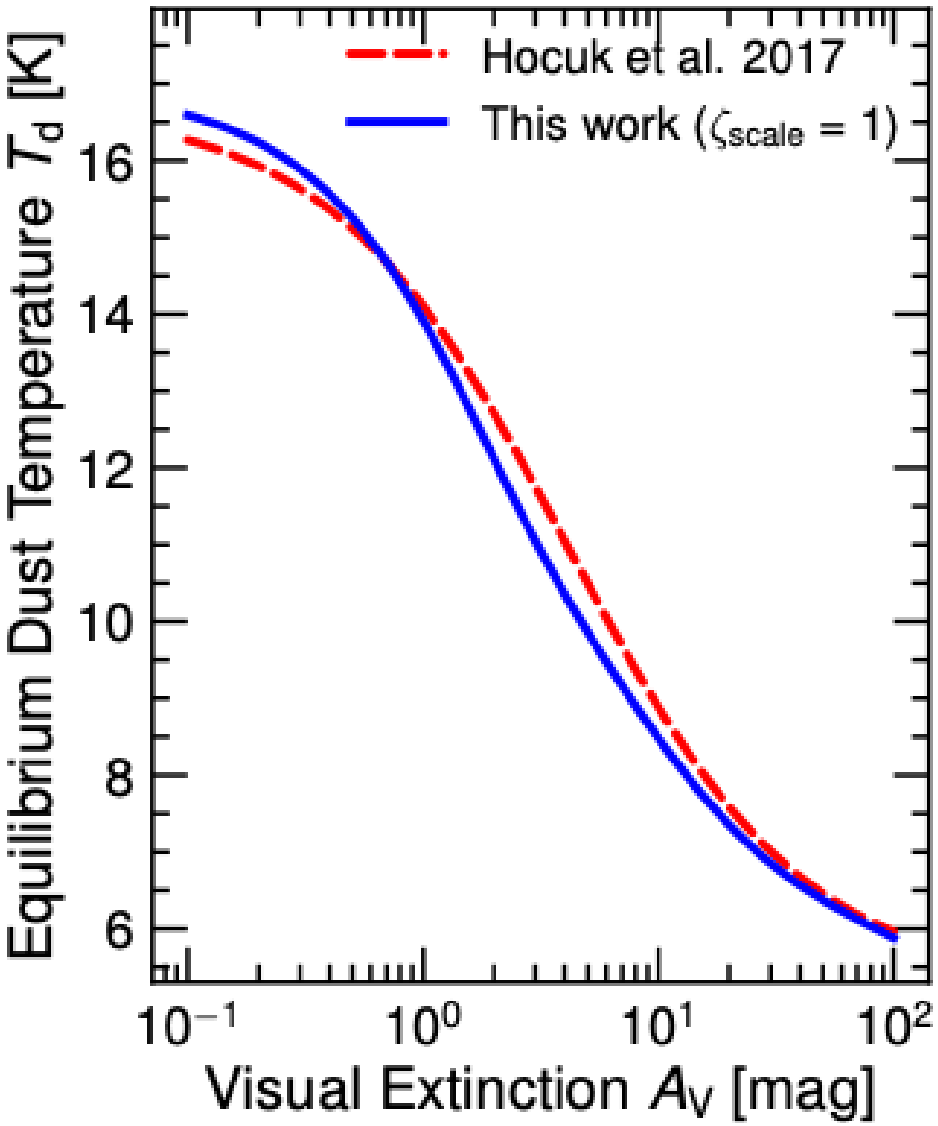
- Optical properties of the bare—core olivine grain
- ISRF spectral density in Draine's unit
- CCM extinction curve (A_λ/A_V)

❑ Compute T_{dust} as a function of Av

Some Hints:

- q_{abs} ~ 0.13 $\mathrm{K^{-2}\ cm^{-2}}$
- $A_\lambda = A_\mathrm{V} \times (A_\lambda/A_\mathrm{v})$

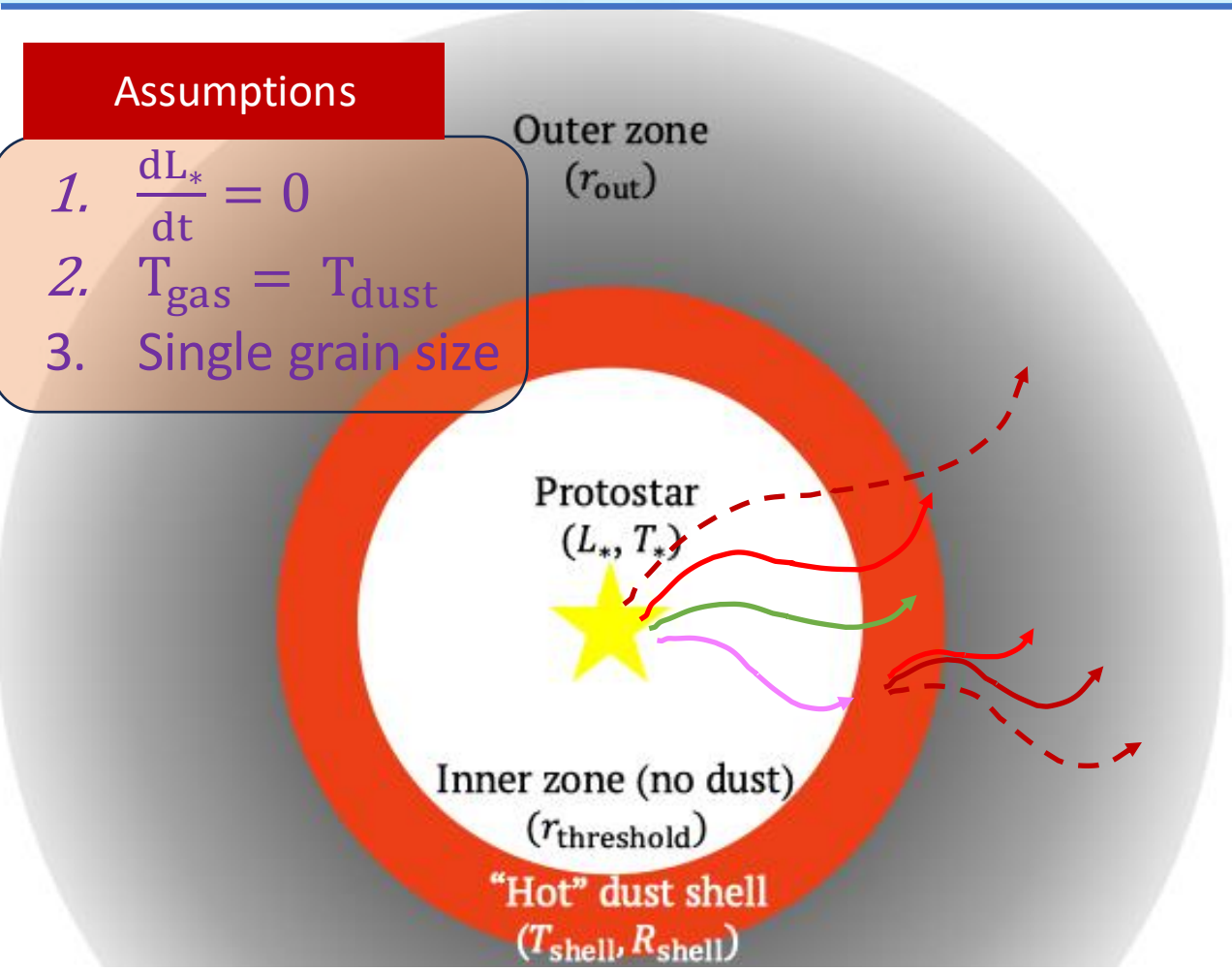
Dust heating by ISRF

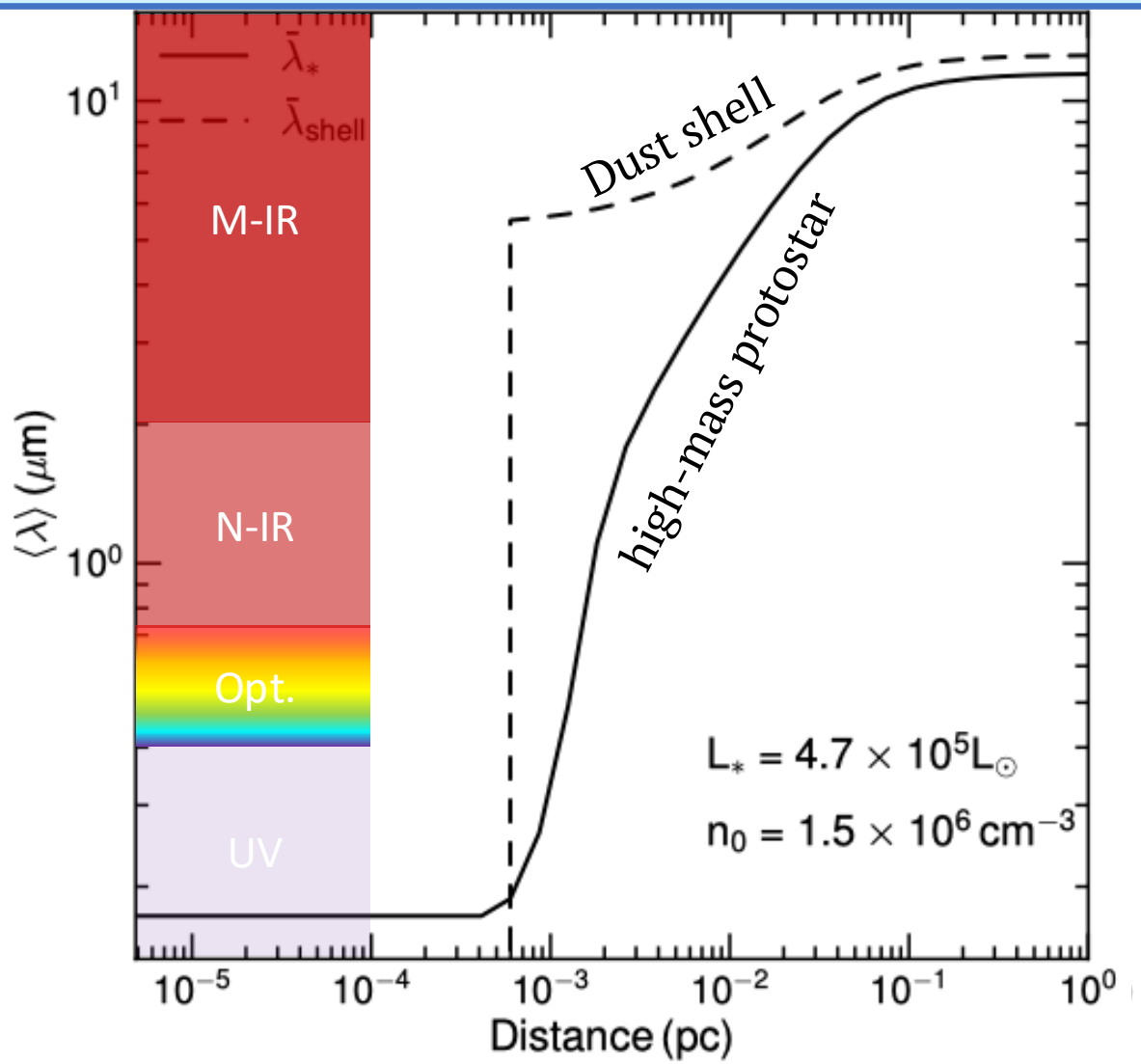


Radiative Treatment in UCLCHEM

Assumptions

1. $\frac{dL_*}{dt} = 0$
2. $T_{gas} = T_{dust}$
3. Single grain size





- **Total radiation**: $U = \int_{13.6\text{ev}}^{20\mu m} \frac{L_\lambda}{4\pi r^2 c u_{ISRF}} e^{-\tau_\lambda} d\lambda$
- **UV-radiation**: G_0^{ext} and G_0^{int}

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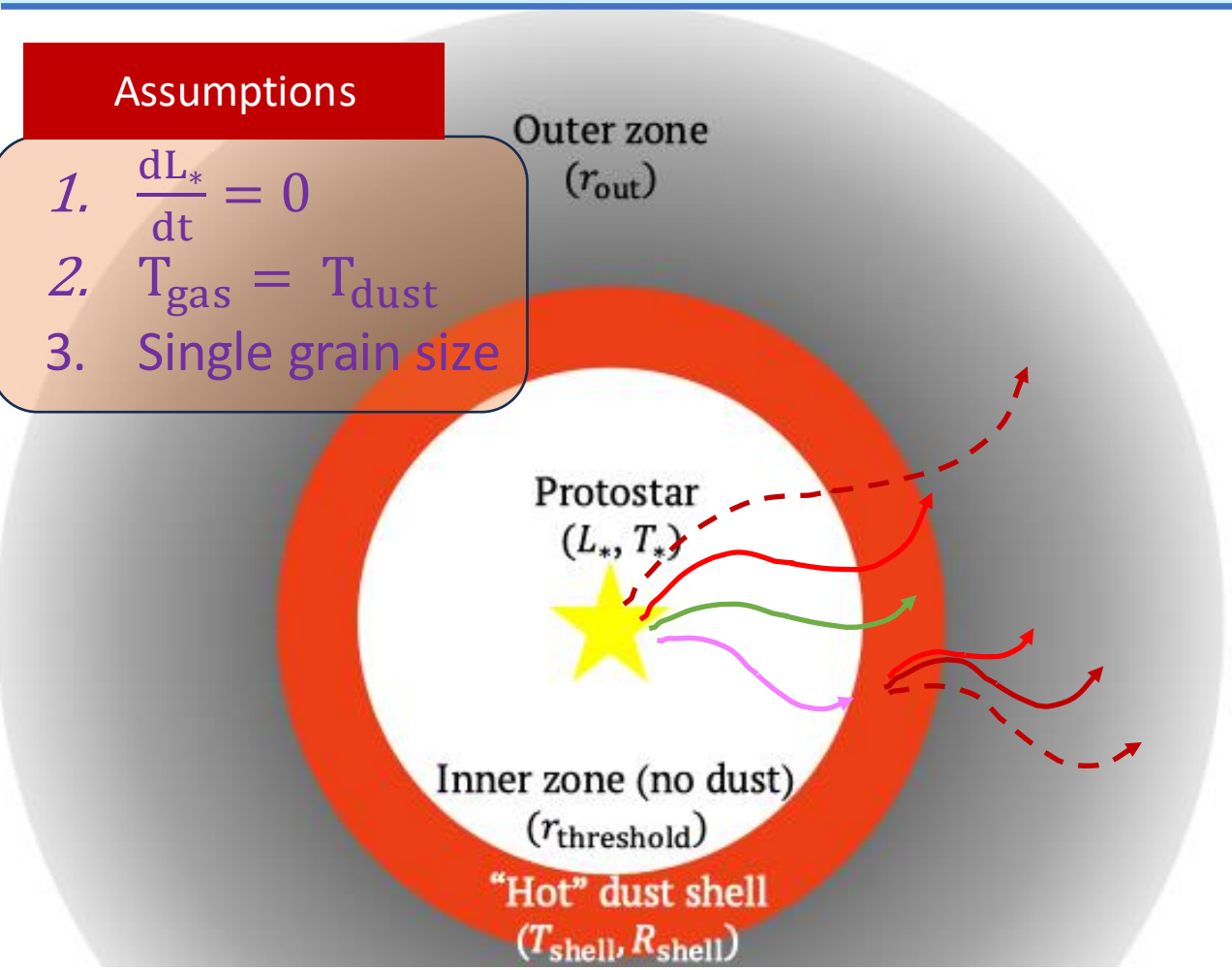

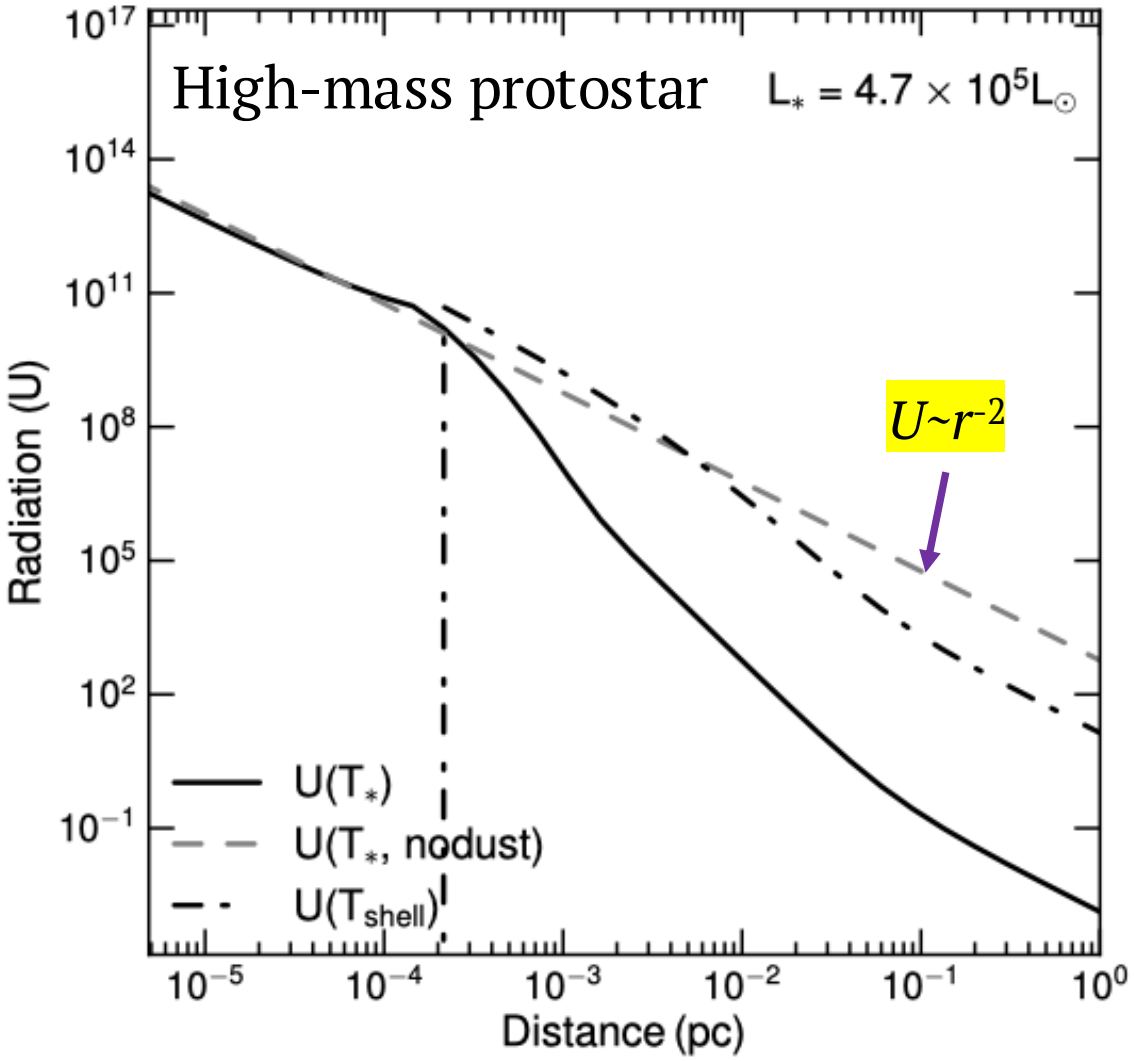

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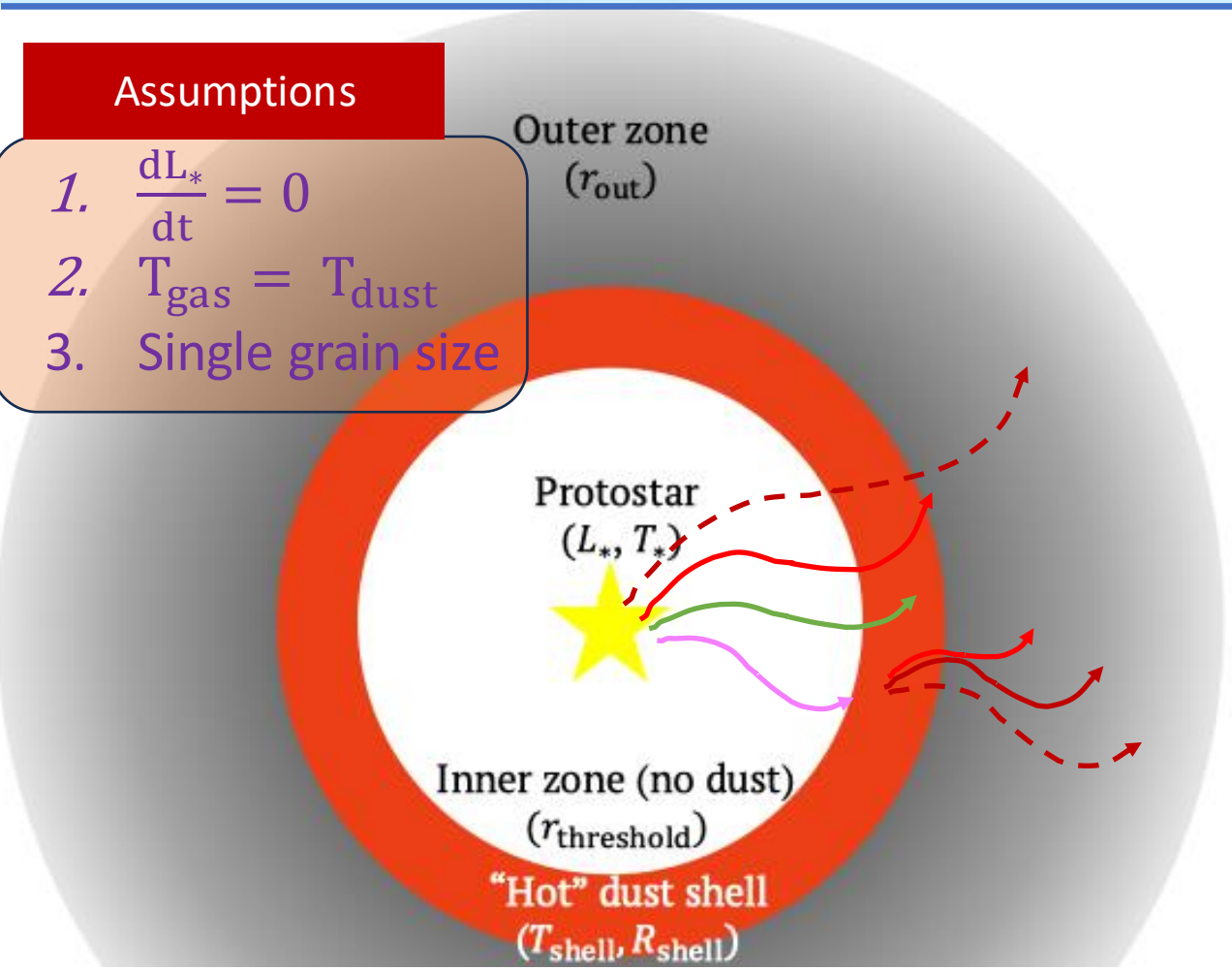

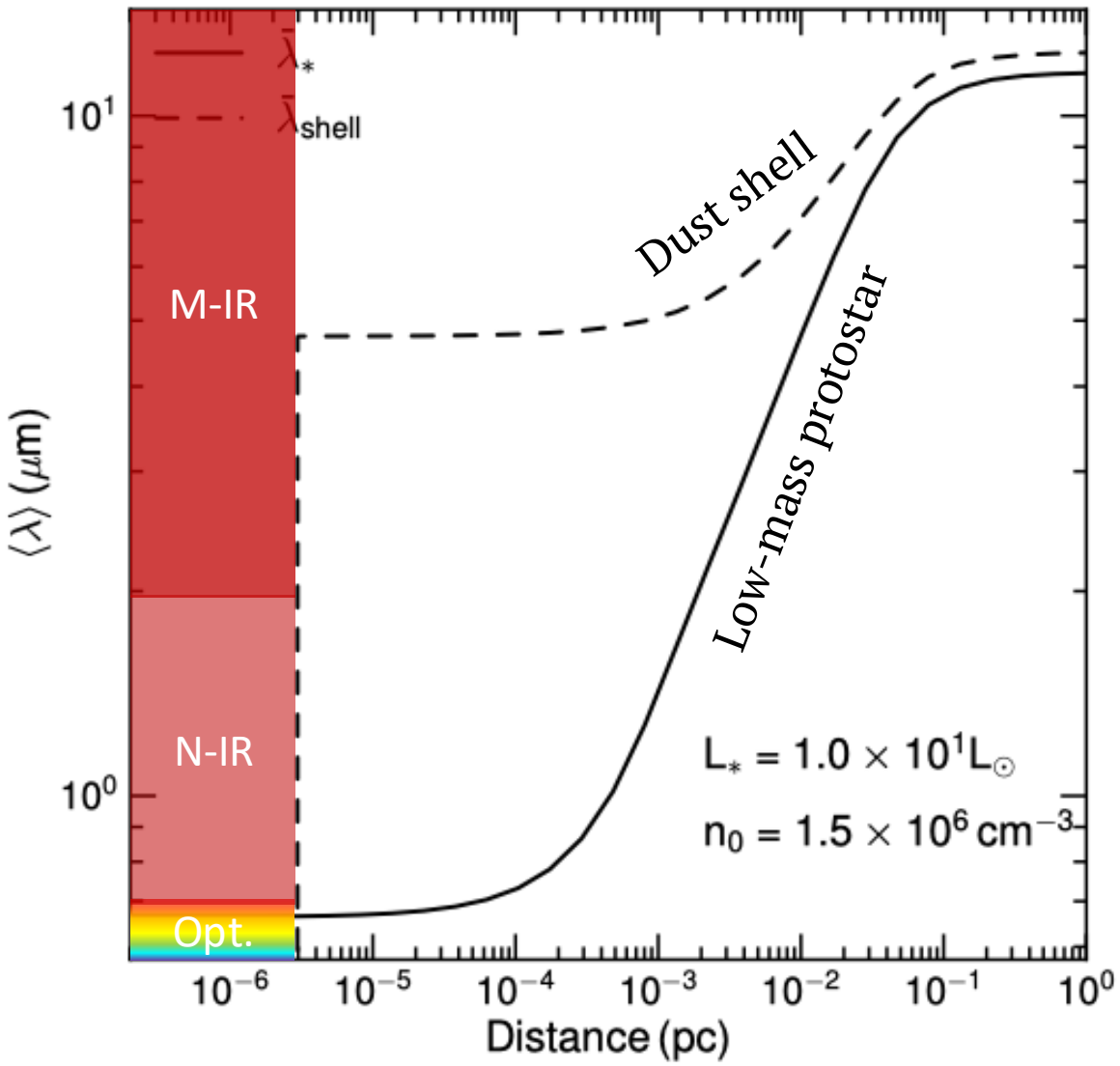

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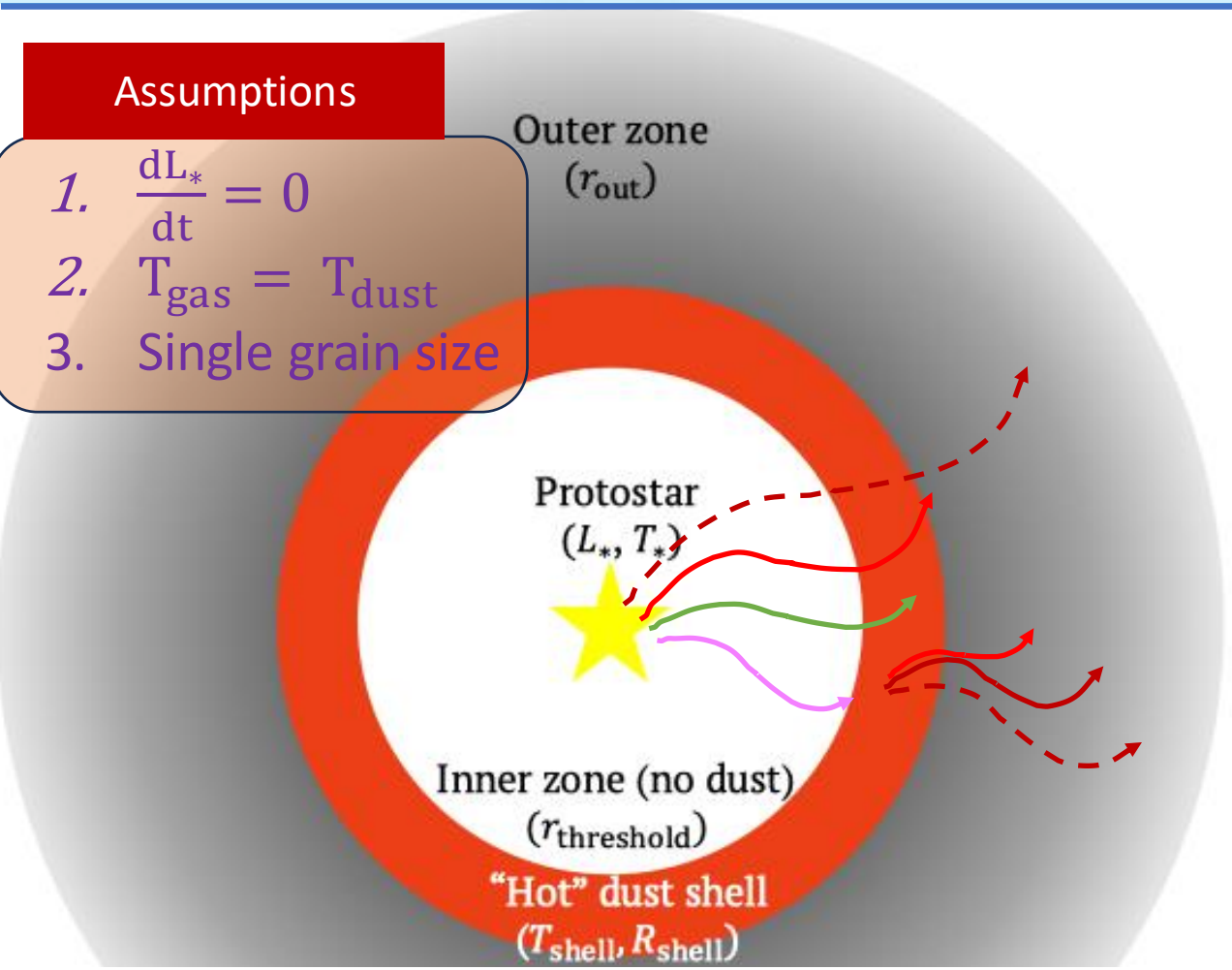
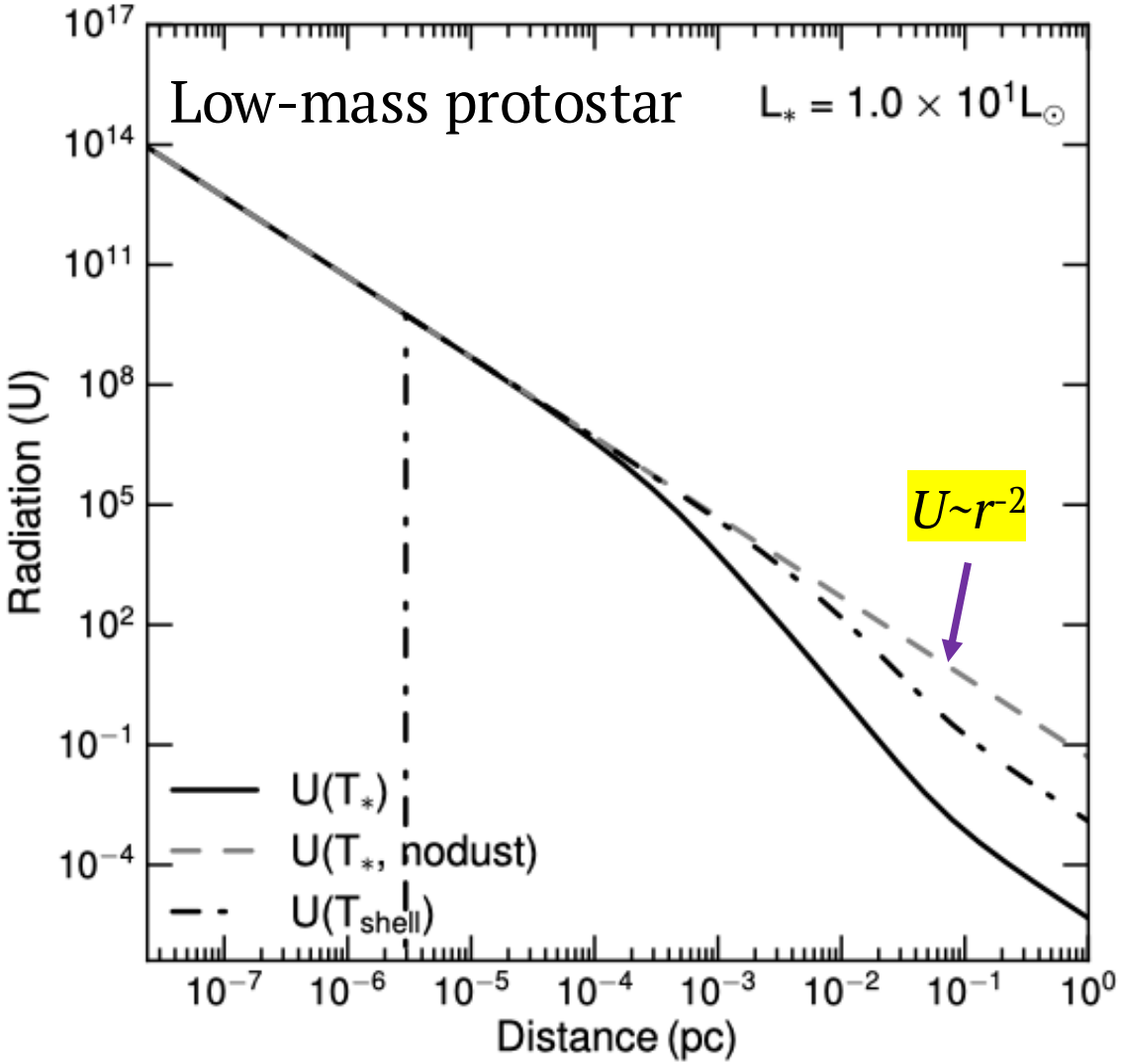
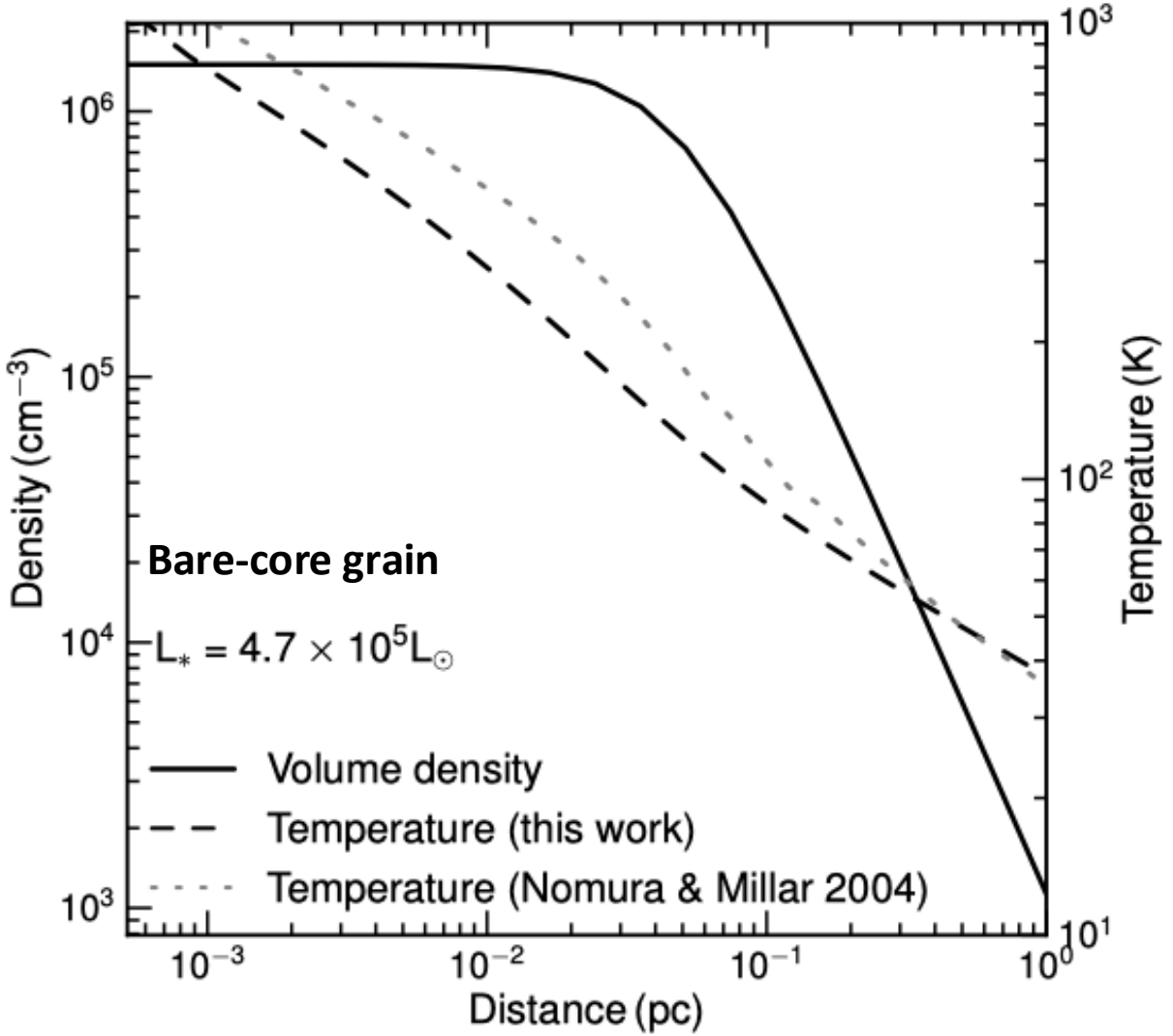
1. $\frac{dL_*}{dt} = 0$
2. $T_{gas} = T_{dust}$
3. Single grain size

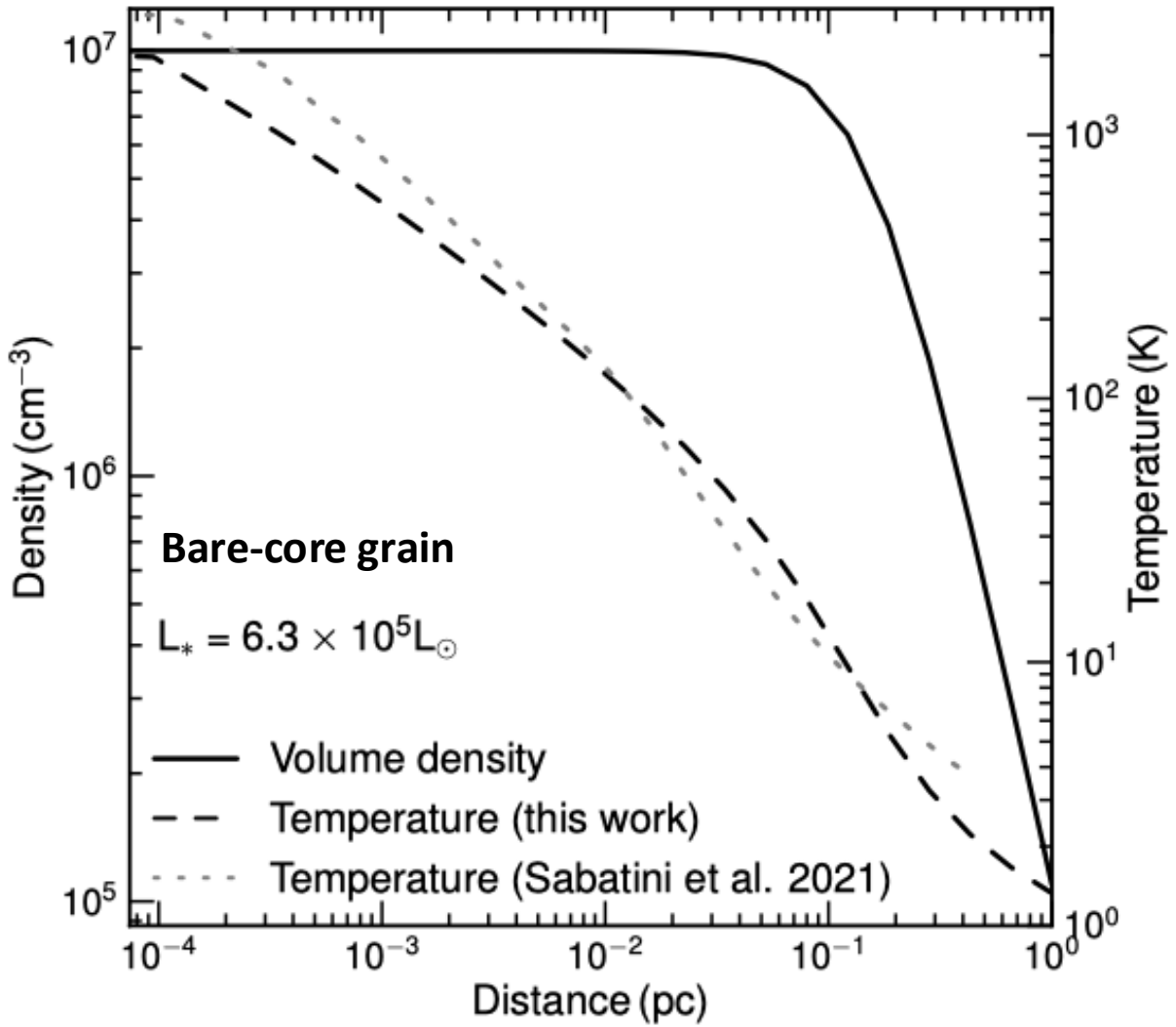




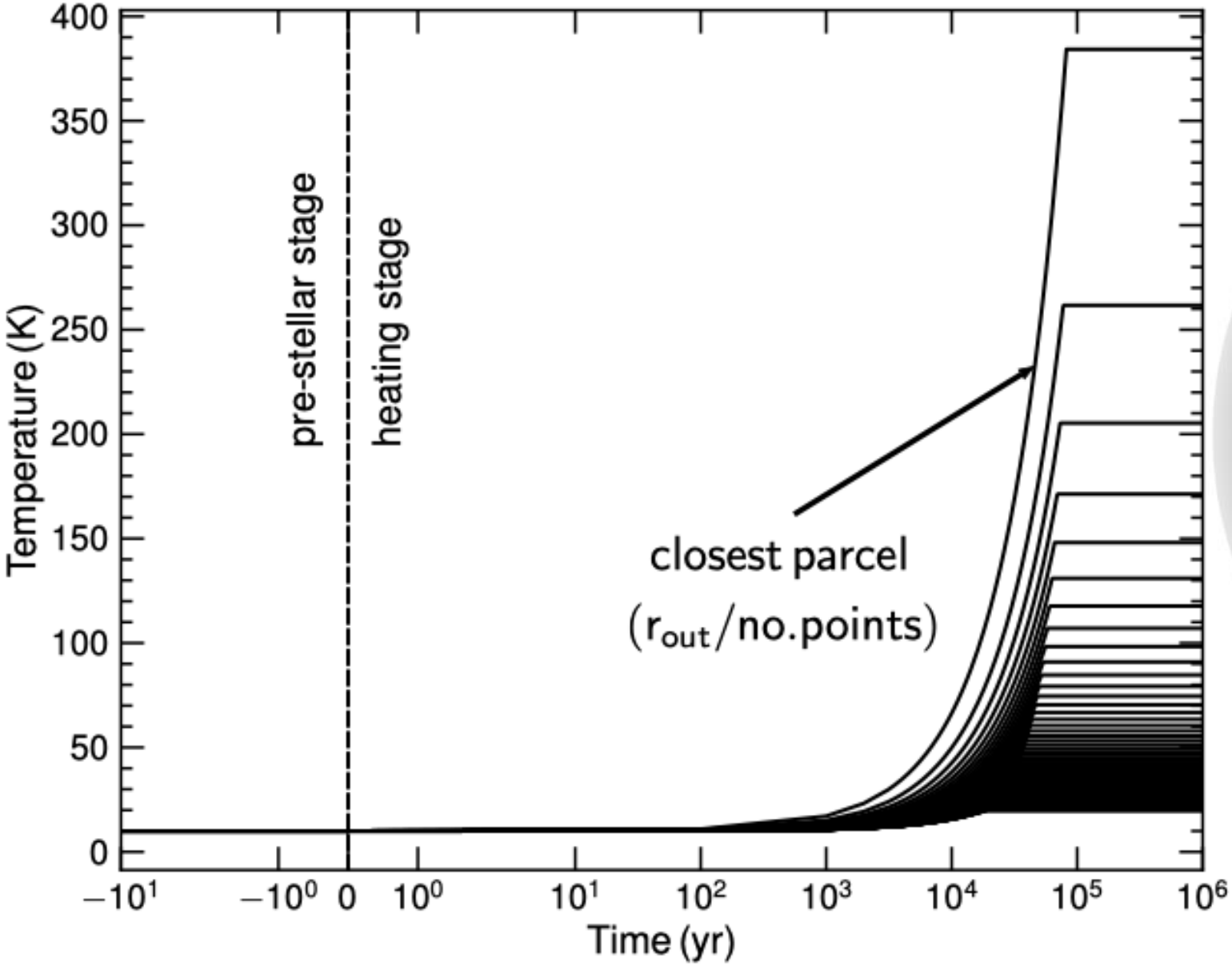
- **Total radiation**: $U = \int_{13.6\text{ev}}^{20\mu m} \frac{L_\lambda}{4\pi r^2 c u_{ISRF}} e^{-\tau_\lambda} d\lambda$
- **UV-radiation**: G_0^{ext} and G_0^{int}

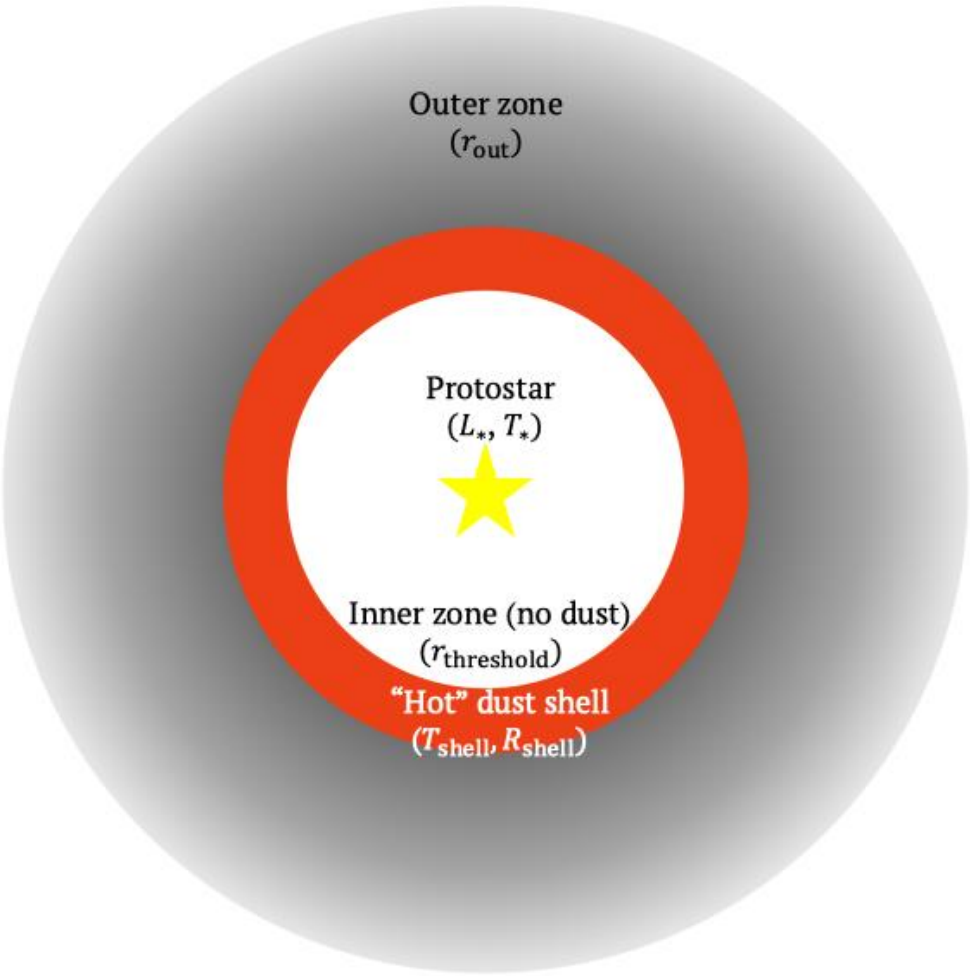
The *radial*-dependent physical profiles



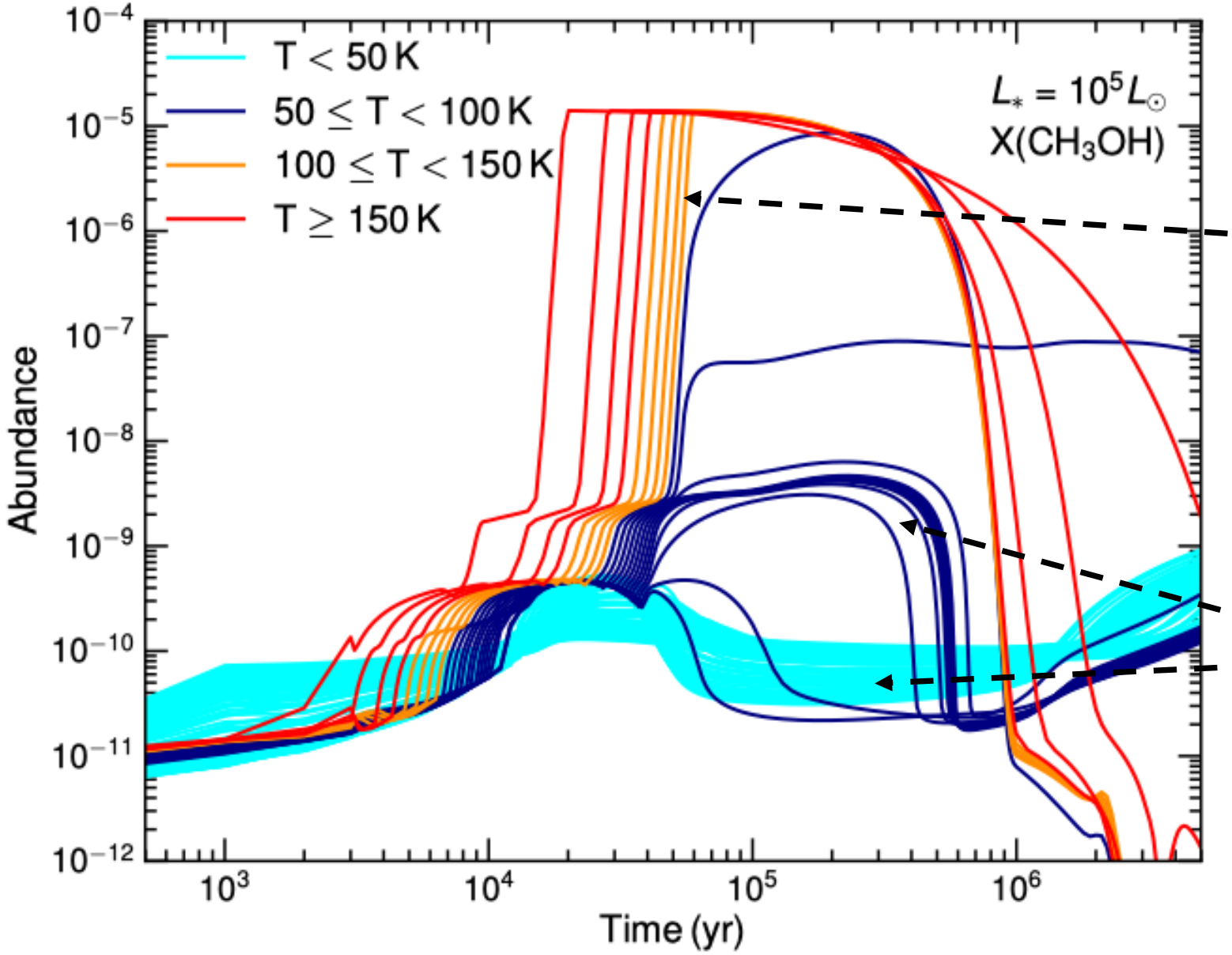


The *radial*- and *time*-dependent physical profiles





The *radial*- and *time*-dependent chemical profiles

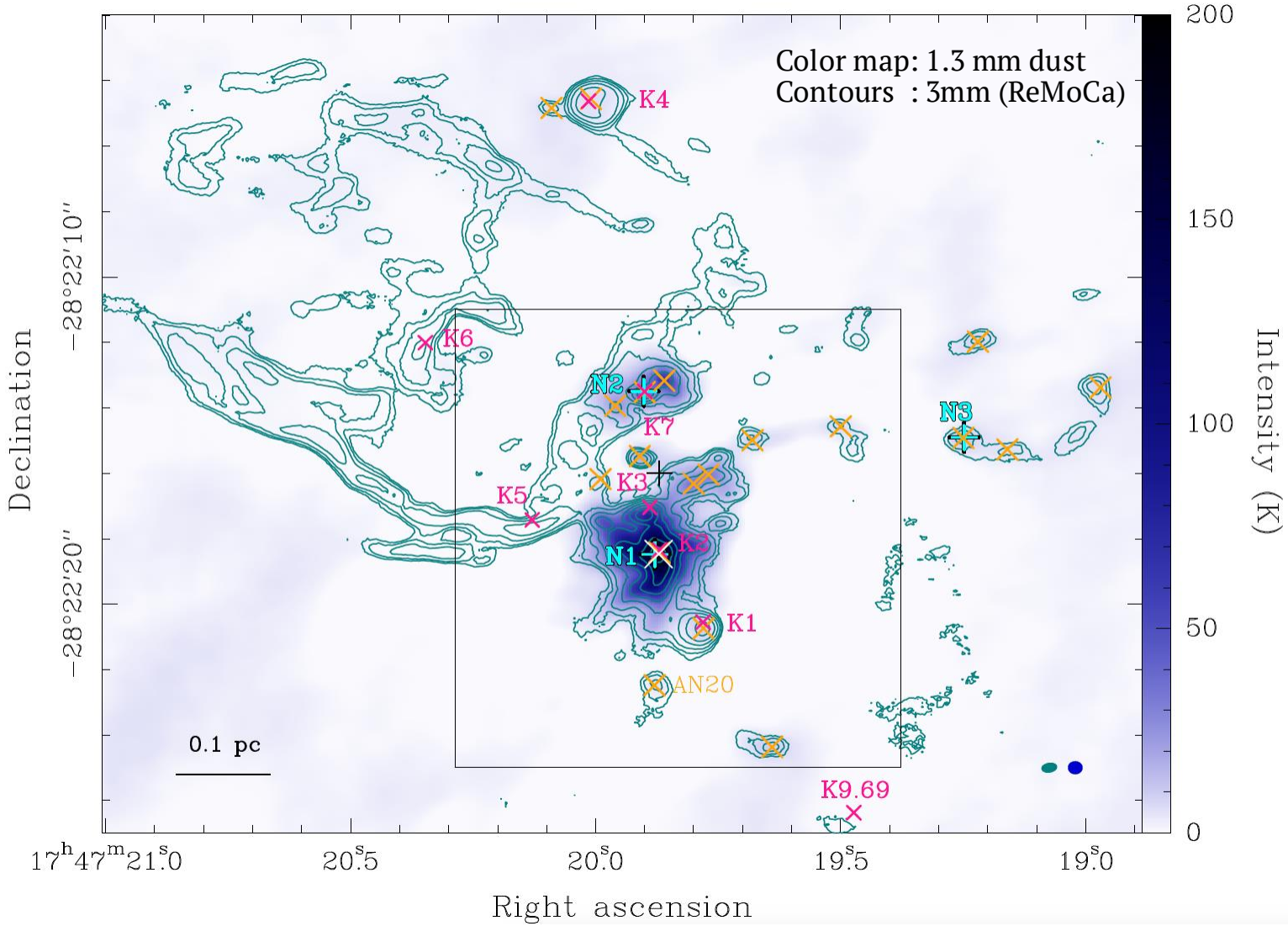


Hot core/Hot corino features

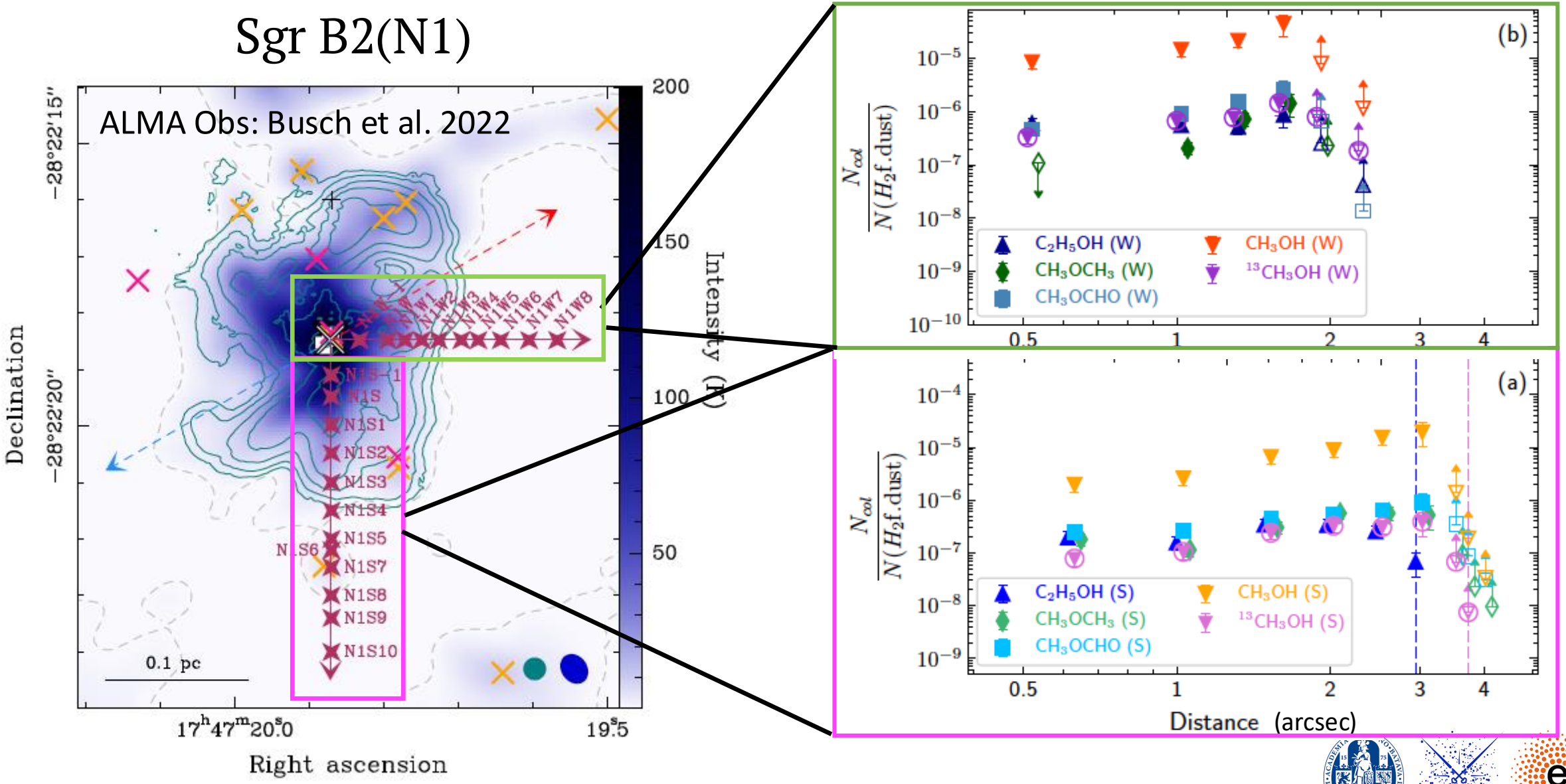
"Cloud"-like features

0D → 1D UCLCHEM model: How important?

Sgr B2 system
(ALMA ReMoCA survey, Belloche et al. 2019)

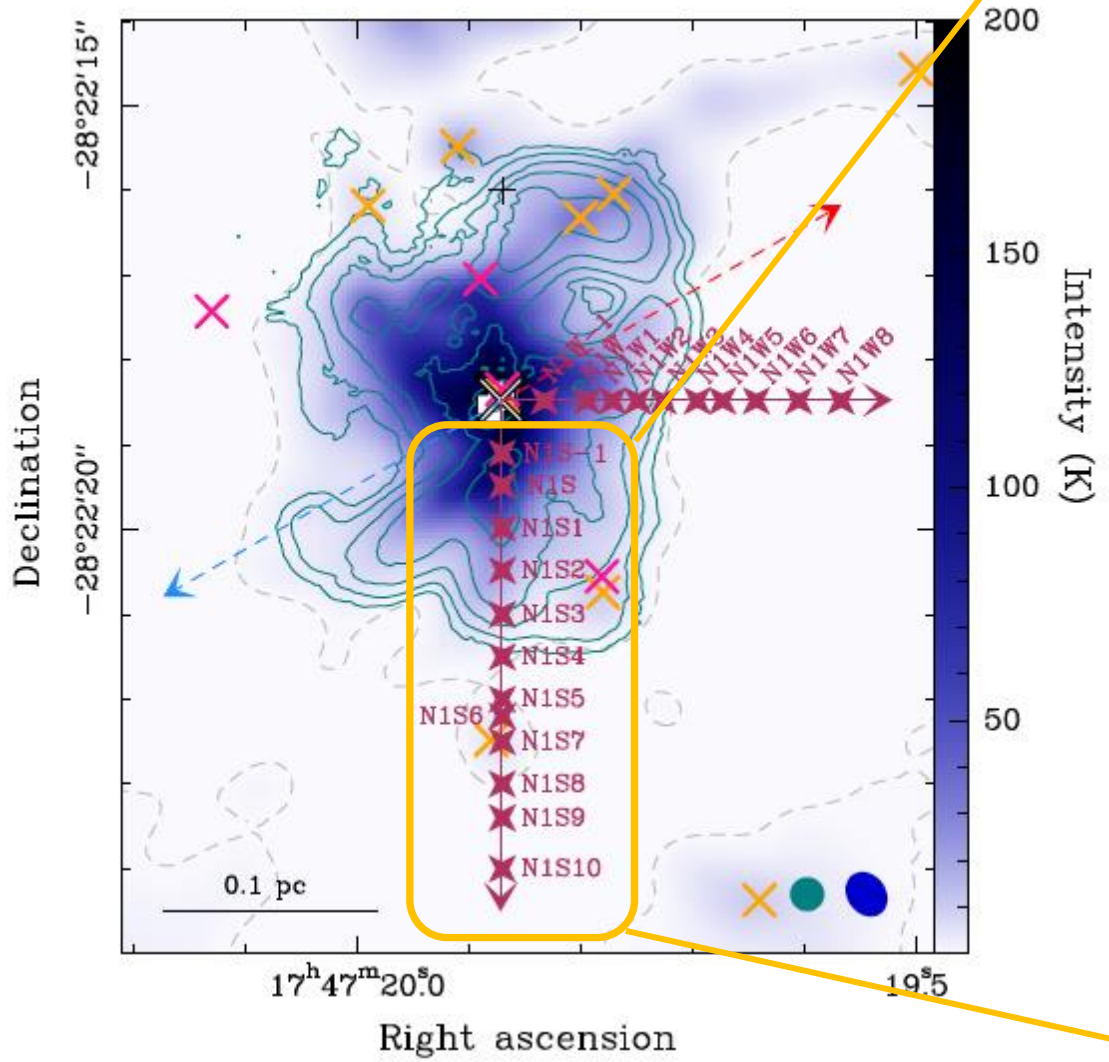


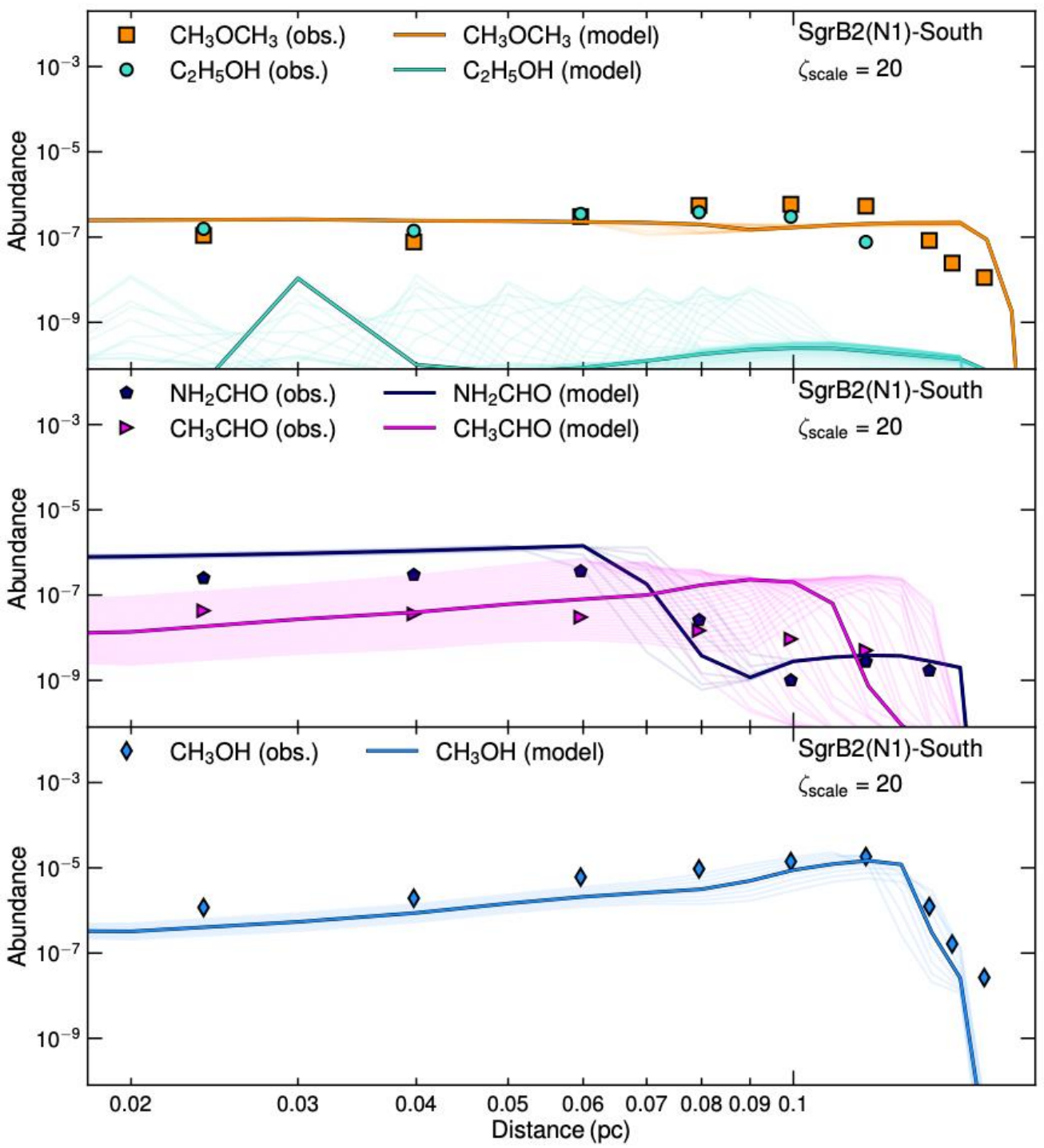
0D → 1D UCLCHEM model: How important?



Sgr B2(N1) hotcore

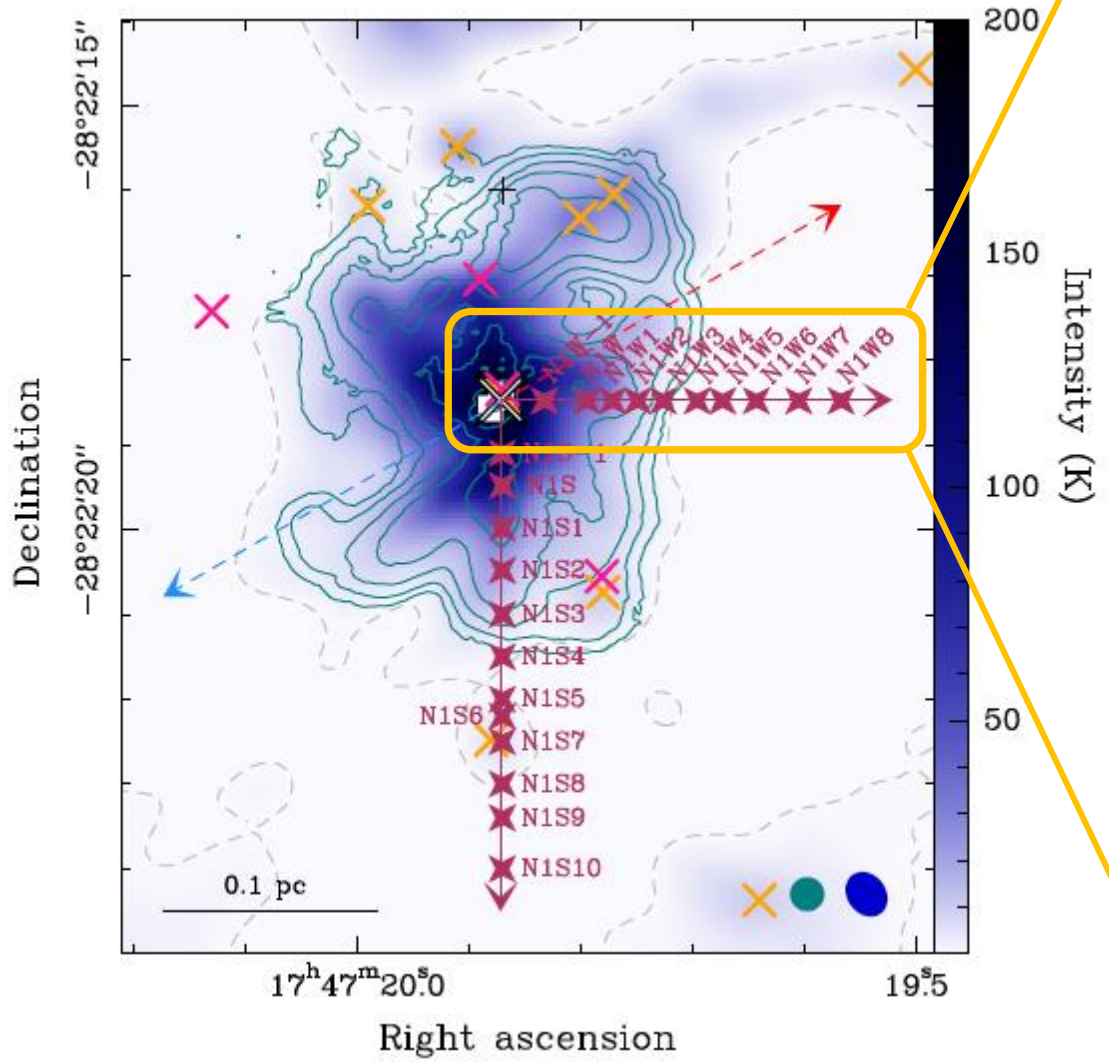
Southward-split

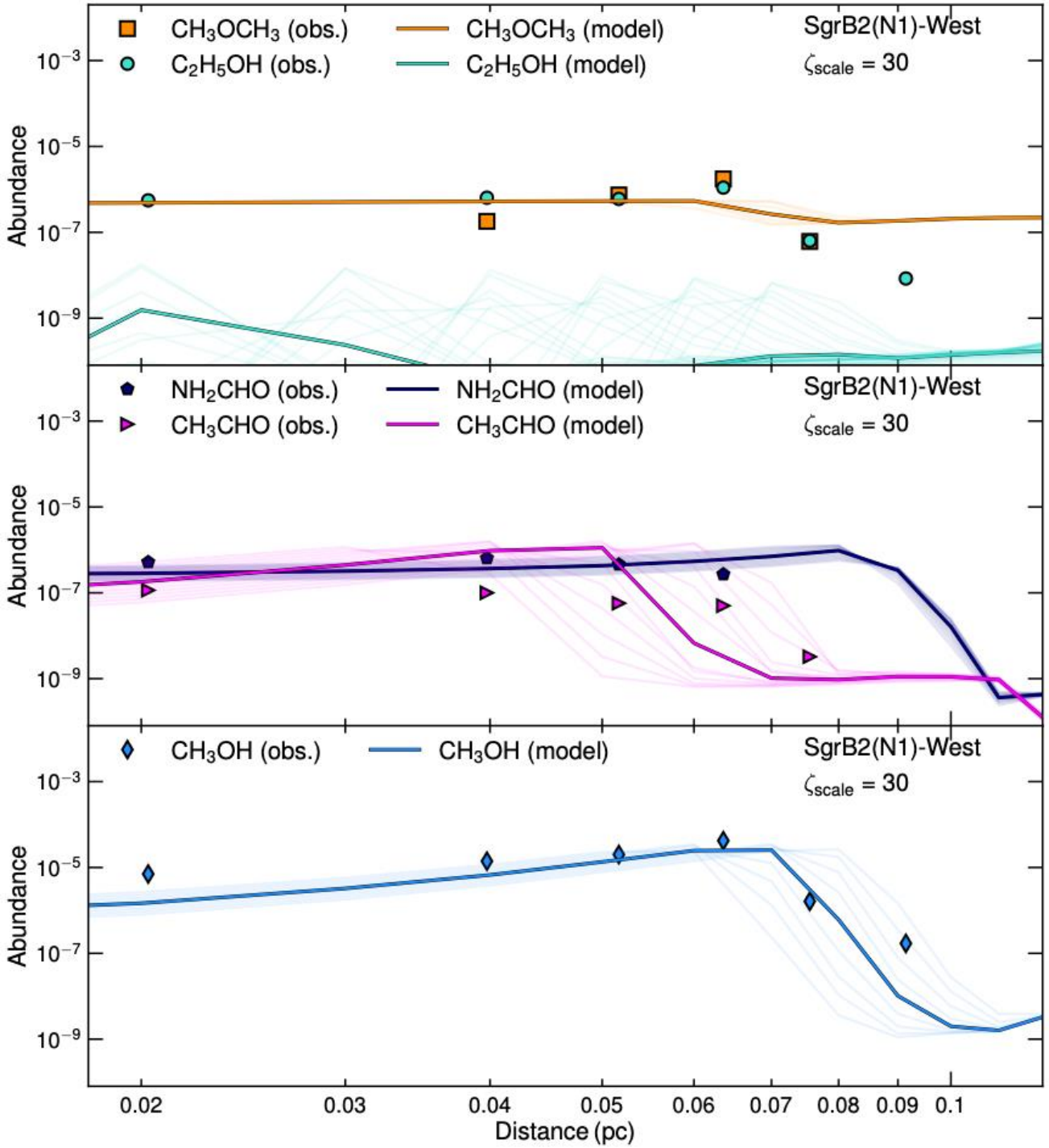




COMs in Sgr B2(N1)

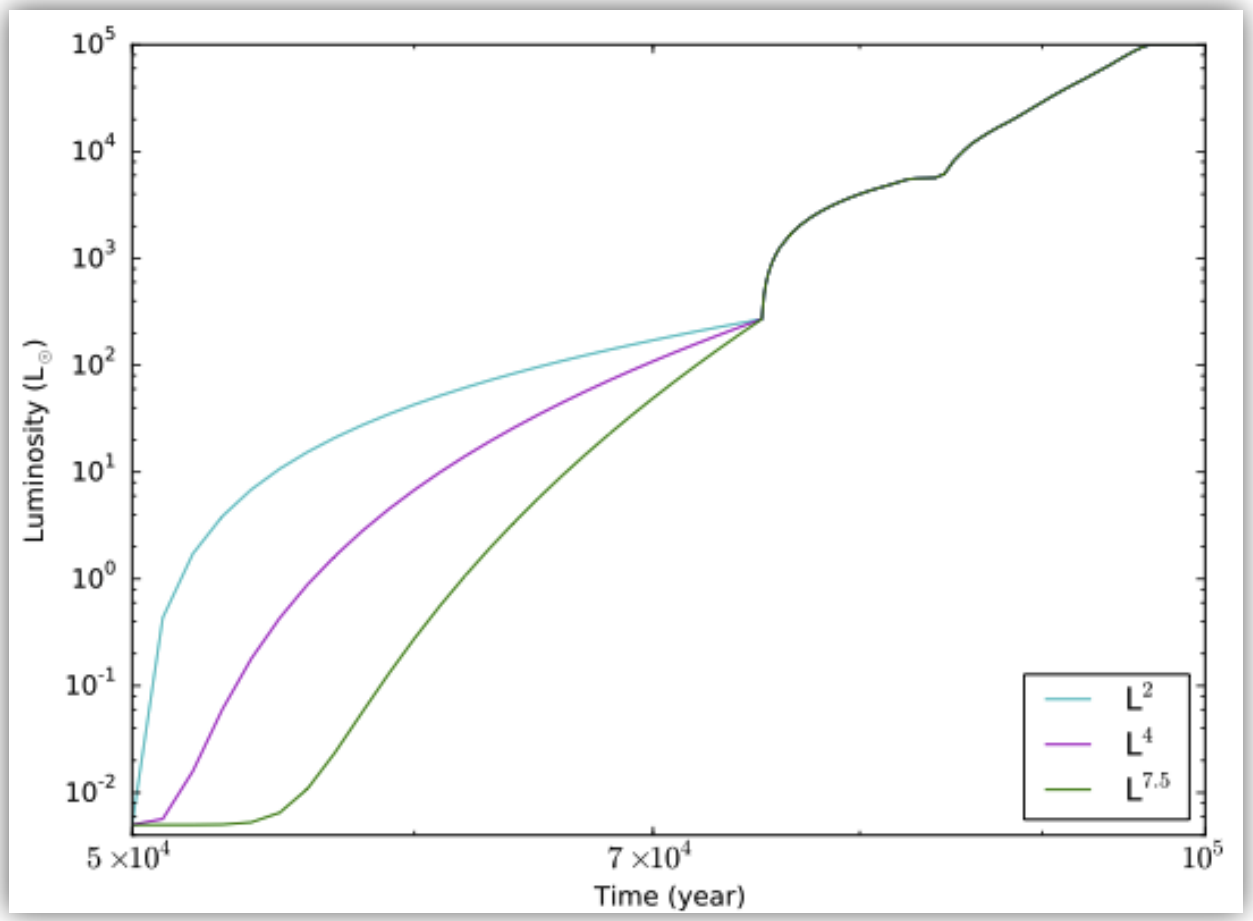
Westward-split





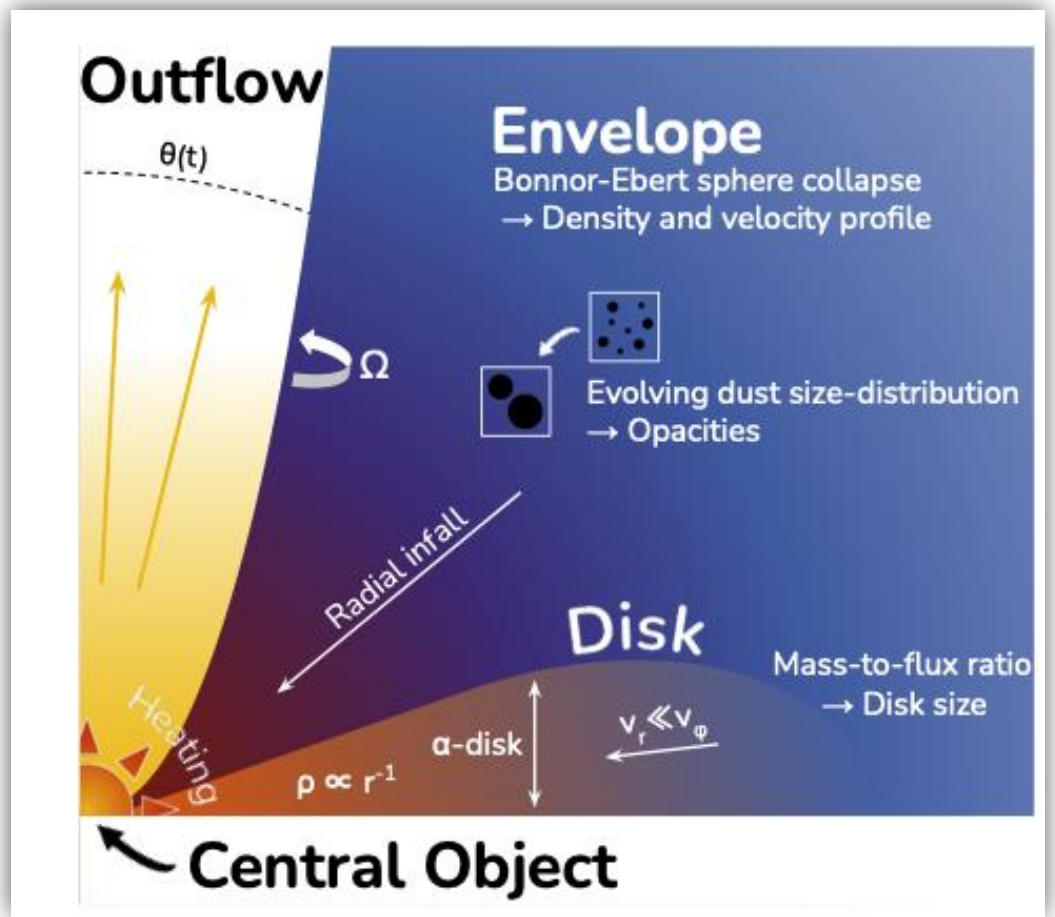
Evol. of L_* and Complex Structure of a Protostar

Parameterisation of luminosity



Choudhury et al. 2015

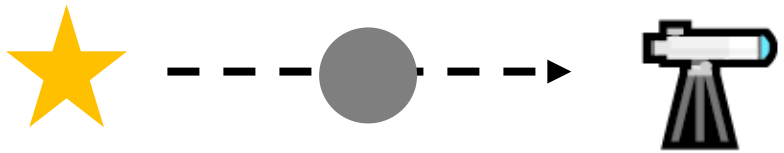
Parameterisation of protostar



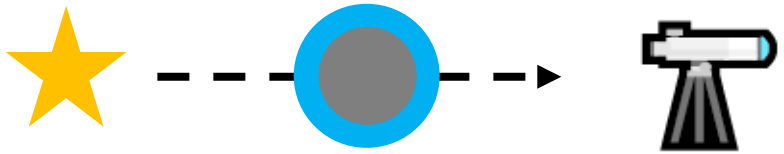
Marchand et al. 2025

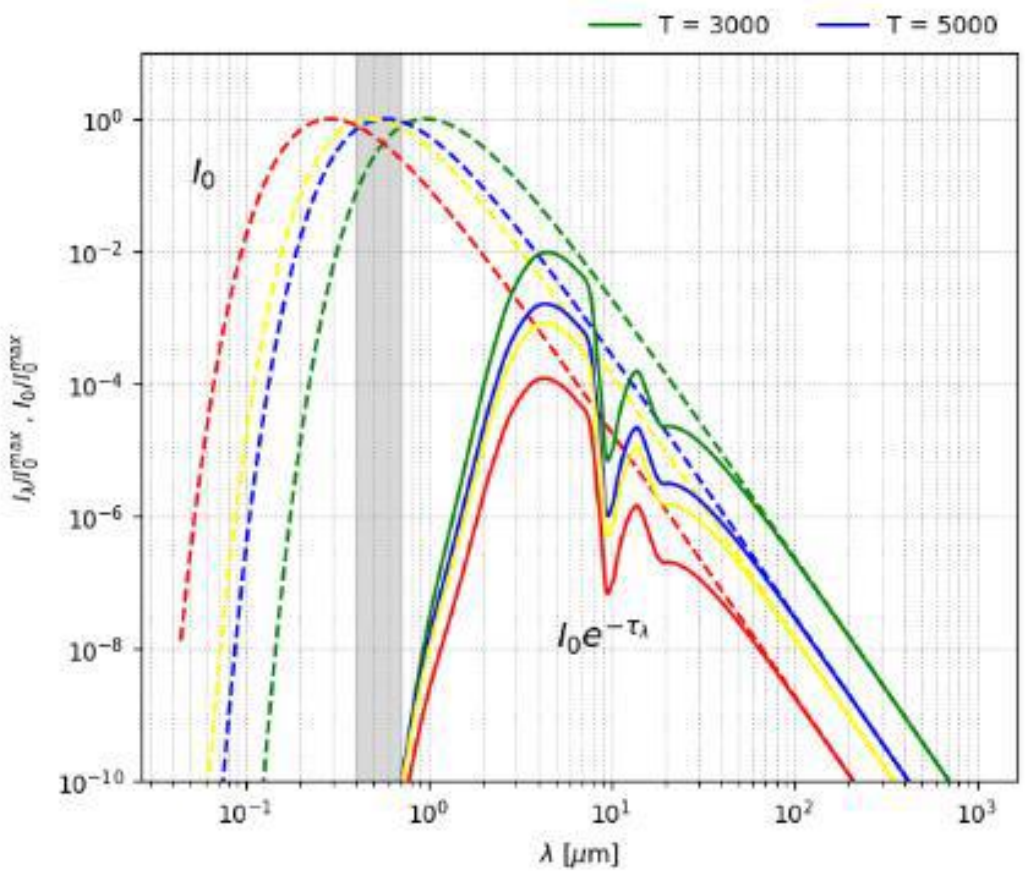
Core-mantle Dust

bare-core dust

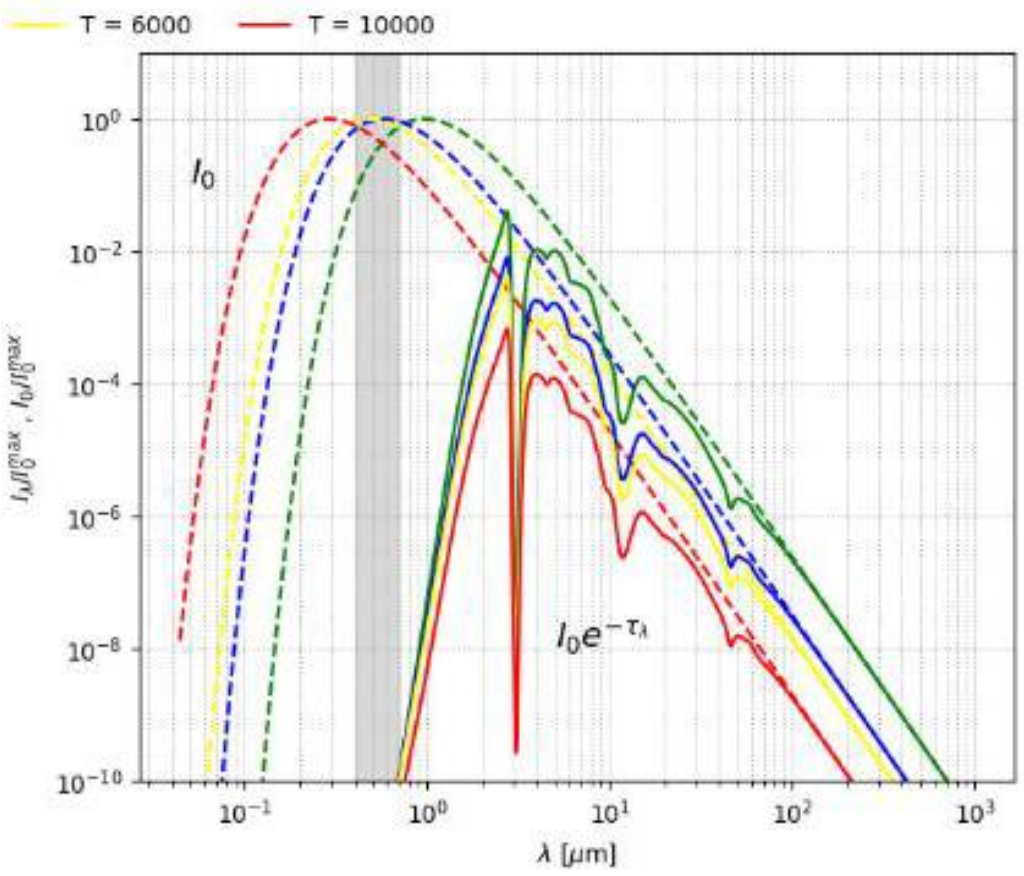


core-ice dust



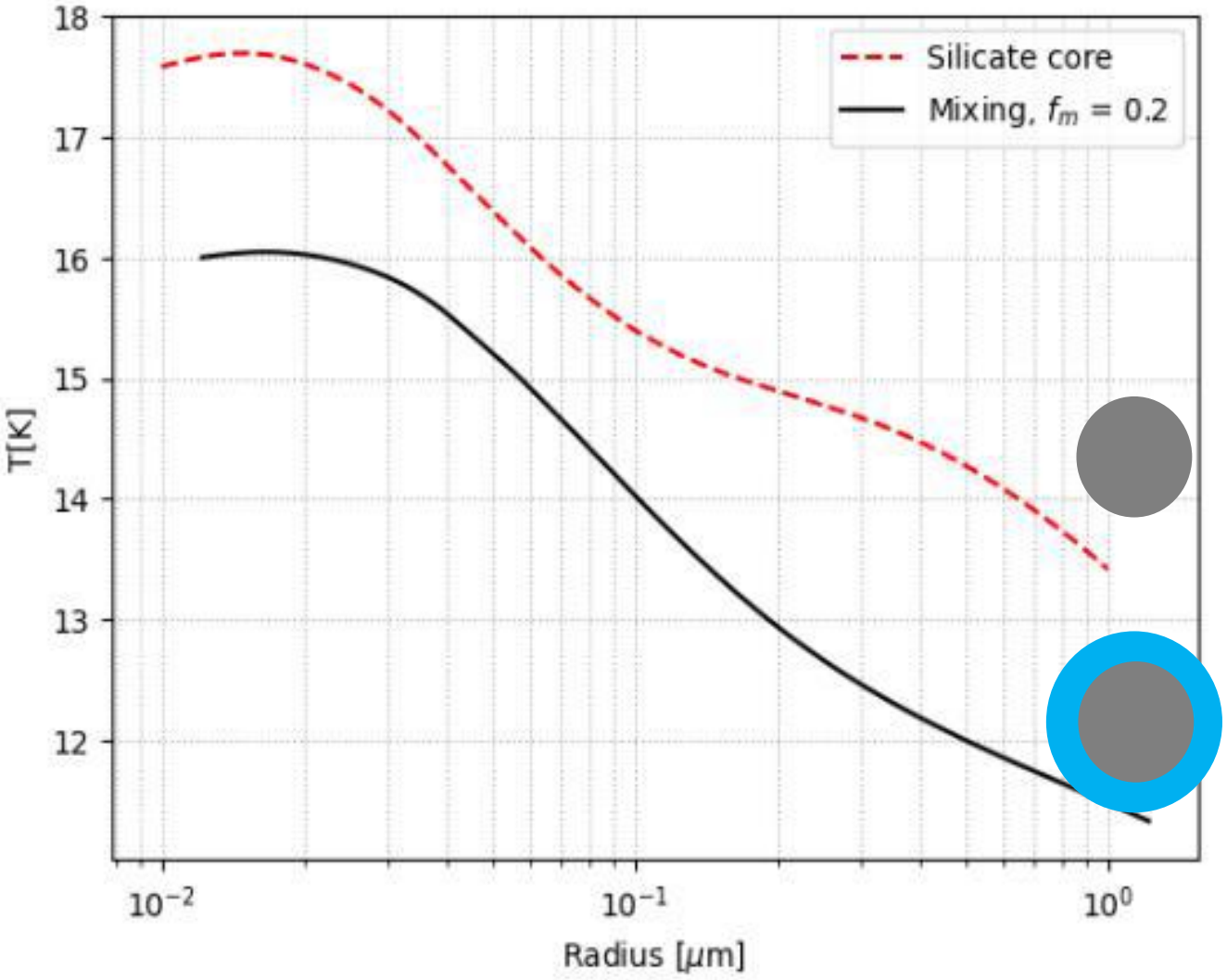


a) Silicate core



b) Mixing, $f_m = 0.2$

Core-mantle Dust



Other physics …

- Radiation field as a quantized quantity? (my afternoon talk)
- Gas-grain temperature decoupling (G. Vermariën's talk)
- Grain growth
- Size distribution
- Grain dynamics
- Clumpy structure of the envelope
- …

Thank you very much for your attention!